

Searching Binary Embeddings

Methods, Memory, Query Time, and Recall

Detailed study, 11 September 2026

Abstract

This study compares cluster-based full scanning, inverted bitplane search, and weighted key lookup for binary document embeddings. The methods share a floating-point query, a binary-document score, and a common reference ranking. The analysis derives their query-time work, index storage, temporary memory requirements, and recall under stated probability models. Small examples explain the data structures, stopping rules, and assumptions before the general calculations. Experiments include independent real queries, controlled query-weight distributions, checks of predicted operation counts, and tests of complete query-time estimates at larger collection sizes. Neither alternative establishes a speed advantage over the tuned scan on the evaluated real embeddings. Controlled inputs identify conditions favoring each alternative while exposing overlap between direct routing and key lookup. Estimates at a billion documents remain conditional on the workload, memory layout, and measured cost model. Failed approximations and unsuccessful settings are retained to define the limits of the conclusions.

Keywords: vector search; binary quantization; indexing; recall; memory. The shared representation and scoring choices follow Exa’s published vector-database description [1]. This report evaluates local implementations; it does not measure Exa’s production system. The shorter formal companion presents the central methods and results.

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1 Scoring

Consider a database containing N documents and a query that requires K results. The objective is to minimize search latency subject to the available RAM and a specified recall target. Comparing algorithms requires a common definition of the reference K documents.

The initial reference ranking consists of the top K documents under the same **float-query, binary-document inner product**. This definition permits comparison of the search indexes under a common score. A later reranking step may use longer float embeddings, but that is a different score and a different reference ranking. An exact search under the binary score can return a different order from float scoring. Neither score necessarily matches a human relevance ranking.

1.1 Storage, memory, and work

1. **Index storage:** data built in advance and kept between queries. Examples are packed document codes, cluster offsets, centroids, bitplanes, and key directories.
2. **Query RAM:** temporary live data needed by one query. Examples are a query vector, score tables, the best R results (stored in a heap), masks for pending branches, and states used to list keys in score order.
3. **Query work:** operations and data accesses over time. Reading the same array a thousand times does not require a thousand stored copies, but it can cost a thousand reads.

The number of scored documents alone does not measure work. A method can score fewer documents and still read more memory or manage a much larger queue.

1.2 Search pipeline

All three main methods begin with clusters, or groups of documents. The no-clustering experiment is the special case $C = P = 1$, not a different definition of the task.

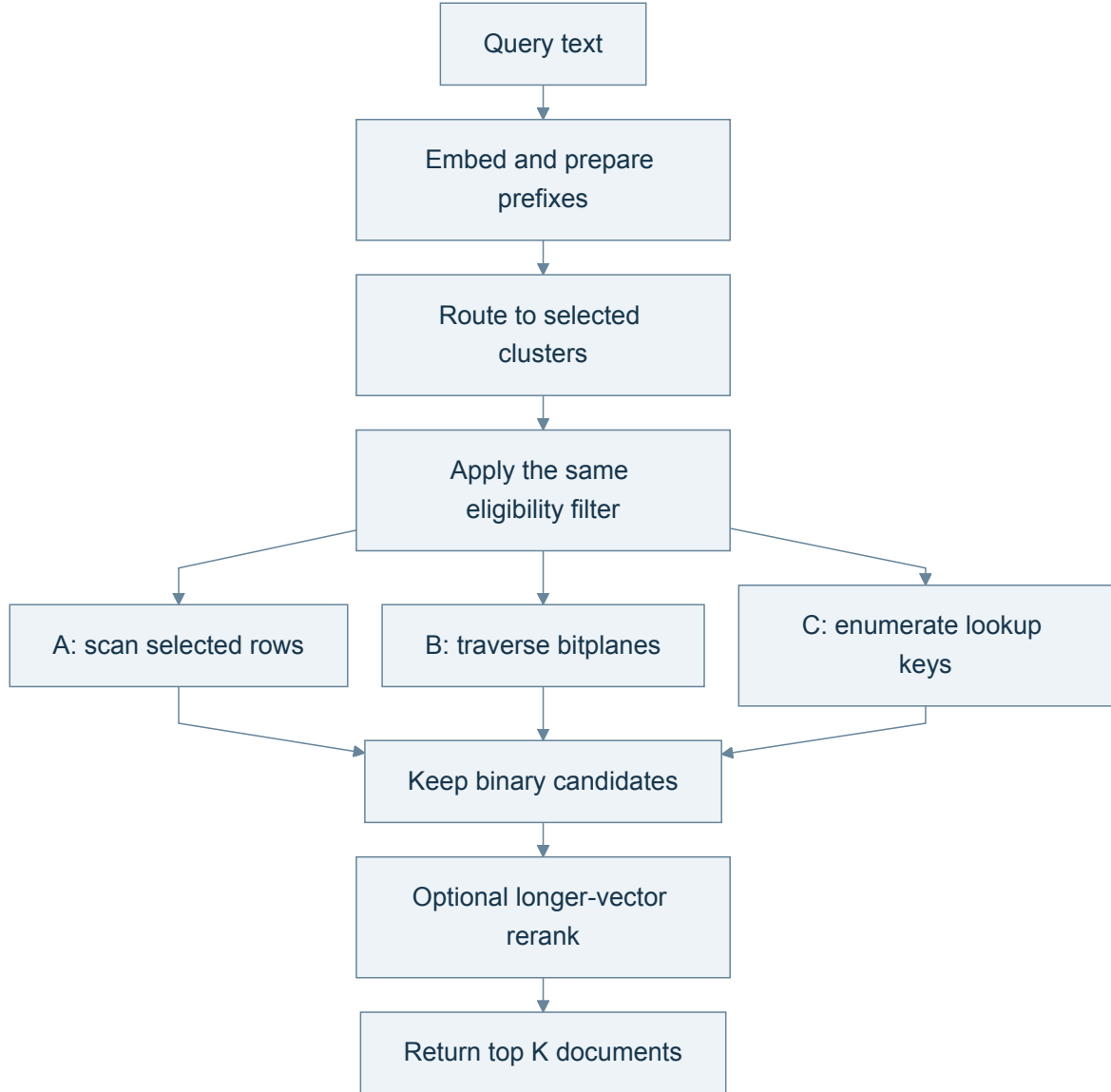


Figure 1: The common query pipeline. Only the candidate-search stage changes between A, B and C.

An embedding model converts query text to a float vector. A short part of the vector selects the clusters to search. Optional filters select eligible documents. The index produces up to R binary-score candidates, with $R \geq K$. An optional final stage fetches longer vectors and scores the candidates again with those vectors, then returns K results.

An index-only experiment uses a cached query embedding and omits the optional reranker. An end-to-end latency measurement includes these stages. The formulas distinguish both measurement scopes.

1.3 Four-bit example

Let

$$q = (0.7, -0.4, 0.2, -0.1), \quad t = \text{sign}(q) = (+1, -1, +1, -1).$$

A stored $+1$ is represented by bit 1 and -1 by bit 0. The ideal bit pattern is therefore 1010. Define $a_i = |q_i|$ and $A = \sum_i a_i = 1.4$. A document bit that agrees with the query contributes $+a_i$; a disagreement contributes $-a_i$.

$$S(q, b) = \sum_{i=1}^d q_i b_i = A - 2 \underbrace{\sum_{i=1}^d a_i \mathbf{1}[b_i \neq t_i]}_{\pi_q(b)}. \quad (1)$$

The quantity $\pi_q(b)$ is the **weighted mismatch penalty**: the sum of query magnitudes at coordinates where the document has the opposite sign. The highest score has the lowest penalty. The factor two comes from replacing a contribution $+a_i$ by $-a_i$, which changes the score by $2a_i$.

Document	Code	Mismatching coordinates	Score
1	1010	none	1.4
2	1011	4	1.2
3	1000	3	1.0
4	0010	1	0.0
5	1110	2	0.6
6	0101	1, 2, 3, 4	-1.4
7	1001	3, 4	0.8
8	0011	1, 4	-0.2

For example, document 7 has penalty $0.2 + 0.1 = 0.3$, so its score is $1.4 - 2(0.3) = 0.8$. The exact global top two are documents 1 and 2. Equal scores are resolved by a stable numeric document ID throughout. This is the internal ID used by the index; an external identifier or URL can be recovered through metadata.

The query defines a preferred sign and a mismatch cost at each coordinate. It does *not* determine whether a preferred pattern exists in the database. This distinction is relevant to both branching and key enumeration.

1.4 Matryoshka prefixes

Exa uses Matryoshka prefixes in its public vector database design [1]. A Matryoshka model is trained so that selected prefixes—the first coordinates of an embedding—remain useful representations [2]. It does not impose $|q_1| \geq |q_2| \geq \dots \geq |q_D|$ for every query. A later coordinate may have a larger magnitude than an earlier one. The training method alone also does not determine the numerical recall of the first h binary signs.

There are three distinct choices:

- The first h coordinates form a fixed prefix that can be indexed in advance.
- Sorting all coordinates by $|q_i|$ produces a query-dependent traversal order suitable for bitplanes.

- A longer representation for final scoring can change the ranking after retrieval.

A hash table built for one fixed prefix does not become a table for the query’s largest-magnitude coordinates when the query is sorted. Such a change would require different lookup coordinates.

1.5 Parameters and units

The formulas below are not tied to a particular code length, cluster count, word width, or SIMD implementation. Table 1 defines the common notation. Additional symbols are introduced where their algorithm first needs them.

Table 1: Shared parameters. Counts and byte sizes are different types of quantity.

Symbol	Meaning	Units
N	Number of stored documents	documents
D	Full embedding dimension	coordinates
d	Dimension of the binary score in Equation (1)	bits / doc
d_r, d_f	Routing and optional final-reranking dimensions	coordinates
C	Number of disjoint clusters	clusters
n_j	Physical number of document rows in cluster j ; $\sum_j n_j = N$	documents
$\mathcal{P}(q), P$	Probed cluster set and its size	set, count
e_j	Eligible rows in probed cluster j , after filters; $e_j \leq n_j$	documents
$N_{\text{sel}}, N_{\text{elig}}$	$\sum_{j \in \mathcal{P}} n_j$ and $\sum_{j \in \mathcal{P}} e_j$	documents
K, R	Final result count and binary candidate-heap capacity, $K \leq R$	documents
w, b_w	Bits and bytes per bitmap word; $b_w = w/8$	bits, bytes
g, G	Bits per score-table group and $G = \lceil d/g \rceil$ groups	bits, groups
b_s, b_q	Bytes per stored table score and query float	bytes
b_{id}, b_o	Bytes per document ID and offset	bytes
b_c, b_f	Bytes per routing-centroid and final-vector coordinate	bytes
M_{RAM}	Available physical RAM budget	bytes
J	Number of simultaneously active queries	queries
c_x	Effective time for a named unit of work x	seconds/unit
$T_{\text{emb}}, Q_{\text{emb}}$	Query-embedding latency and its peak scratch RAM	seconds, bytes

Assume positive integer dimensions and counts, $1 \leq P \leq C$, $1 \leq K \leq R$, and a word width w divisible by eight. A selected pool may contain fewer than R eligible rows, in which case fewer candidates are returned.

The following widths recur in the storage and work formulas:

$$s_{\text{code}} = b_w \lceil d/w \rceil, \quad W_j = \lceil n_j/w \rceil, \quad s_{\text{mask},j} = b_w W_j. \quad (2)$$

The first describes one packed *document*; the second describes one bitplane across a *cluster*. One stores bits for a document; the other stores a bit for each document in a cluster. If a filter leaves only a few eligible rows, W_j does not shrink unless the data is physically rebuilt or represented differently.

1.6 Claims and evidence

An algebraic identity, such as Equation (1), is exact under its stated score. An operation count is exact for the stated pseudocode or layout, but can change with the implementation. A latency or statistical estimate also needs hardware or distribution assumptions. Measured tables report a particular experiment. These categories distinguish formulas and estimates from measured results.

2 Shared pipeline

2.1 Exa’s design

Exa’s public vector database post describes Matryoshka prefixes, binary quantization of documents, float queries scored against those binary documents, and lookup tables for groups of four coordinates. It also describes similarity-based clustering, filtering, and final reranking [1]. Its published example uses particular dimensions and cluster counts; the formulas in this report replace those choices with parameters. The pseudocode here is an explicit reference design, not a claim to reproduce Exa’s private implementation or production latency.

2.2 Index construction

A document is encoded before queries arrive. Its code and document ID are placed in its cluster’s contiguous row range. In the baseline layout, each document has one packed code and one ID. Cluster offsets identify the ranges. Bitplanes and lists of documents for each hash key, called postings, are extra structures described later.

Listing 1: Common preprocessing; executed before serving queries.

```
BuildCommonIndex(documents, model, d, routing_configuration):
    vectors = encode documents with model
    routing = train or choose the cluster assignment rule
    for each document x:
        b = signs of the first d coordinates of vectors[x]
        j = assign x to one cluster
        append (document_id[x], packed(b)) to cluster j
    concatenate cluster rows and record C + 1 offsets
    save centroids if centroid routing is used
    save filter metadata and optional final vectors
    return common index
```

Encoding, clustering and sorting happen before queries arrive, so their time belongs to index construction. The data they produce counts toward index storage. Build-time temporary arrays can be large; the serving-RAM formulas below deliberately do not describe the construction peak.

For centroid routing, define

$$M_{\text{base}} = N s_{\text{code}} + N b_{\text{id}} + (C + 1) b_o + C d_r b_c + M_{\text{filter}} + M_{\text{extra}}. \quad (3)$$

M_{extra} includes explicitly chosen common metadata and optional stored final vectors. For example, reranking vectors kept entirely in RAM contribute $N d_f b_f$. If they are on disk, they still consume

storage, but only the portion kept in RAM belongs in serving RAM. For sign-prefix routing, omit the centroid term. Comparisons must count ID arrays consistently across methods, including any additional copies.

Example: Stored data

Suppose an illustrative database has $N = 10^9$ documents and $d = 256$ bits. The packed code data, without extra padding, is $10^9 \cdot 256/8 = 32$ GB. With eight-byte IDs, the codes and IDs together need 40 GB before centroids or metadata. Here GB means 10^9 bytes. These are example choices for N, d, b_{id} , not constants in Equation (3).

2.3 Query preparation

Listing 2: Query preparation shared by all methods.

```
PrepareQuery(text, model, D, d, d_r):
    full_query = model.encode(text, D)
    q = prepare prefix full_query[0:d] for the chosen score
    routing_query = prepare prefix full_query[0:d_r]
    t[i] = +1 if q[i] >= 0 else -1
    a[i] = abs(q[i])
    return full_query, q, routing_query, t, a
```

The model’s tokenization, neural-network computation and output processing cost $T_{\text{emb}}(\ell_q, D)$ for query length ℓ_q . This time requires measurement; it does not follow a universal linear formula in D . Taking prefixes, optional normalization, and finding signs and magnitudes add $T_{\text{prep}} = O(D + d + d_r)$ elementary work. Their coefficient can be measured separately. A norm or layer-normalization rule must be the same rule used when preparing the documents.

Query-vector storage is $O(D + d + d_r)$ floats, with possible views instead of copies. The memory account distinguishes copied buffers from shared data. The actual retained query buffers are denoted by Q_{vec} and the embedding model’s peak temporary memory by Q_{emb} .

2.4 Cluster selection

A flat centroid router scores all C centroids and retains the best P . This is distinct from scoring binary documents: centroids are normally floats.

Listing 3: A simple flat centroid router with bounded selection memory.

```
Route(q_route, centroids, P):
    nearest = heap with capacity P
    for j in all C clusters:
        route_score = dot(q_route, centroid[j])
        offer (route_score, j) to nearest
    return the P selected cluster IDs
```

Let c_{fma} be the time for one centroid coordinate’s multiply-add contribution, and let H_r be the number of heap comparisons or equivalent selection operations. Then

$$T_{\text{route}} = C d_r c_{\text{fma}} + H_r c_{\text{sel}}, \quad H_r = O(C \log(P + 1)). \quad (4)$$

Keeping only the best P centroids in a heap needs $O(P)$ records; storing all centroid scores needs $O(C)$ scores instead. Either is valid, but their RAM formulas differ. Hierarchical or approximate routing has a separately measured visited-centroid count; that count can replace C only when the algorithm avoids the other centroids. Its routing recall also needs evaluation.

A sign-prefix router uses r fixed sign coordinates and at most 2^r addresses. It is not the same as similarity-based centroid clustering. Document assignment has low cost, whereas ranking many possible query buckets requires additional work. One reference router scores all C bucket keys using $\lceil r/g \rceil$ lookup terms each and selects P , giving

$$T_{\text{route,sign}} = T_{\text{table,r}} + C \lceil r/g \rceil c_{\text{lut}} + H_r c_{\text{sel}}.$$

A direct or multi-probe address generator can replace that router, but must count its own key-generation and queue work. When every cluster is probed, routing can simply return all cluster IDs and skip distance-based selection.

2.5 Selected documents

The selected document count is

$$N_{\text{sel}}(q) = \sum_{j \in \mathcal{P}(q)} n_j. \quad (5)$$

Only under balanced clusters is this approximately NP/C . An optional load factor $\lambda(q) = N_{\text{sel}}/(NP/C)$ summarizes deviation for a *given query*; it is not automatically constant as P changes. For bitplanes, the individual n_j matter because wider clusters make each node more expensive. The average does not determine the largest cluster size or the peak memory needed for pending nodes.

2.6 Filters

A filter can return eligible row IDs or a bitmap. Denote its time by T_{filter} and its temporary memory by Q_{filter} . For a bitmap intersection over the probed clusters, a reference work count is $\sum_{j \in \mathcal{P}} W_j$ word visits per input mask. For sorted ID lists, cost depends on list lengths and the intersection method instead.

After filtering, e_j rows are eligible in cluster j , so a scan scores $N_{\text{elig}} = \sum_j e_j$ documents. A dense bitplane still has physical width W_j . A filter that removes most rows therefore helps scoring immediately, but does not automatically make each bitplane intersection short.

The reference set for recall must obey the *same filter*. Otherwise a correct filtered search is penalized for excluding ineligible documents.

2.7 Score tables

Exa’s lookup-table scoring approach [1] partitions the d document signs into groups of g ; Exa’s described choice is $g = 4$. A group has at most 2^g sign patterns. For each query group, the score table stores a precomputed dot product with every possible pattern.

Listing 4: The score table. Padding contributes zero in the last group.

```
BuildScoreTable(q, g):
    for group in ceil(len(q) / g) groups:
        for pattern in all g-bit patterns:
            table[group, pattern] = 0
            for coordinate in this group:
                add q[coordinate] * sign(pattern, coordinate)
    return table

ScoreDocument(packed_code, table):
    score = 0
    for group in all groups:
        pattern = extract this group's bits from packed_code
        score += table[group, pattern]
    return score
```

For this straightforward build,

$$G = \lceil d/g \rceil, \quad U_{\text{build}} \leq G2^g g, \quad (6)$$

$$T_{\text{table}} = U_{\text{build}} c_{\text{build}}, \quad Q_{\text{table}} = G2^g b_s, \quad (7)$$

$$T_{\text{score}}(F) = FGc_{\text{lut}}. \quad (8)$$

c_{lut} includes extracting the pattern, the lookup and accumulation in the chosen scoring routine. Faster table construction is possible, but it must replace U_{build} explicitly. Larger g reduces the number of lookup terms and increases the table exponentially. Register capacity and cache behavior therefore matter.

Example: Two query values and a four-entry table

For the first two values $(0.7, -0.4)$, the contributions for patterns 00, 01, 10, 11 are respectively $-0.3, -1.1, 1.1, 0.3$. With $g = 2$, document 1011 uses the first-group entry 1.1 and the second-group entry 0.1, giving 1.2.

XOR marks differing bits, and a population count counts them. Together they compute *unweighted* Hamming distance, the number of differing positions. It matches Equation (1) only when all query magnitudes are equal up to a common scale. Replacing weighted scoring by that operation changes the task.

2.8 Candidate selection and reranking

A heap stores the best R candidates, with the worst retained candidate at its root. Every scored document needs at least a comparison with that root once the heap is full. Let $H_s(F, R)$ be the actual selection-operation count. A simple worst-case bound is $O(F \log(R + 1))$; selection work must therefore be included.

$$T_{\text{select}} = H_s(F, R)c_{\text{heap}}, \quad Q_{\text{results}} \approx R(b_{\text{id}} + b_s) + Q_{\text{heap,overhead}}. \quad (9)$$

For an optional rerank of R' returned candidates, $K \leq R' \leq R$, let

$$T_{\text{finish}}(R') = T_{\text{fetch}}(R', d_f) + R'd_f c_{\text{fine}} + H_f(R', K)c_{\text{fine,sel}} + T_{\text{output}}(K). \quad (10)$$

T_{fetch} depends on whether vectors are already in RAM, on SSD, or remote. If u_f vectors are fetched in a batch, the temporary float-vector buffer is $u_f d_f b_f$, not necessarily $R' d_f b_f$. The candidate IDs must remain live until fetching and scoring finish.

2.9 Time and memory

For sequential stages, define

$$T_{\text{common}} = T_{\text{emb}} + T_{\text{prep}} + T_{\text{route}} + T_{\text{filter}} + T_{\text{table}}. \quad (11)$$

The method-specific candidate work and Equation (10) are added later. For an index-only measurement, the embedding and final-stage terms are set to zero.

Peak temporary RAM is the most memory held by buffers at the same time. It is not the sum of every temporary array allocated during a query. If embedding, search and reranking do not overlap,

$$Q_{\text{peak}} = \max\{Q_{\text{emb,stage}}, Q_{\text{search,stage}}, Q_{\text{finish,stage}}\}. \quad (12)$$

Each stage term includes buffers retained from previous stages. The term Q_{base} denotes common buffers live during candidate search, including query vectors, score tables, filter masks, routing output, and the candidate heap.

For J concurrent queries sharing one resident index,

$$\begin{aligned} M_{\text{serving}} &= M_{\text{resident}} + M_{\text{model}} + M_{\text{runtime}} + \max_t \sum_{u \in \mathcal{A}(t)} Q_u(t) \\ &\leq M_{\text{resident}} + M_{\text{model}} + M_{\text{runtime}} + JQ_{\text{peak,max}}. \end{aligned} \quad (13)$$

M_{model} counts shared resident embedding-model weights when inference is local. An external embedding service contributes latency but no local model-weight allocation. M_{runtime} is a chosen reserve or measured runtime overhead. It is not another stored copy of the index. Separate process replicas or device copies must be counted if the actual system has them.

For separate CPU and GPU memories, stored data and peak temporary memory are counted in each pool, including any additional device copies. A throughput speedup does not by itself establish a feasible memory layout.

3 Full scan

Approach A is the reference candidate-search method. Once clusters and eligible rows are chosen, it scores every eligible row. There is no approximation *inside* that selected pool. The loop repeats the same operations and can read packed codes in order when filtering does not leave scattered rows.

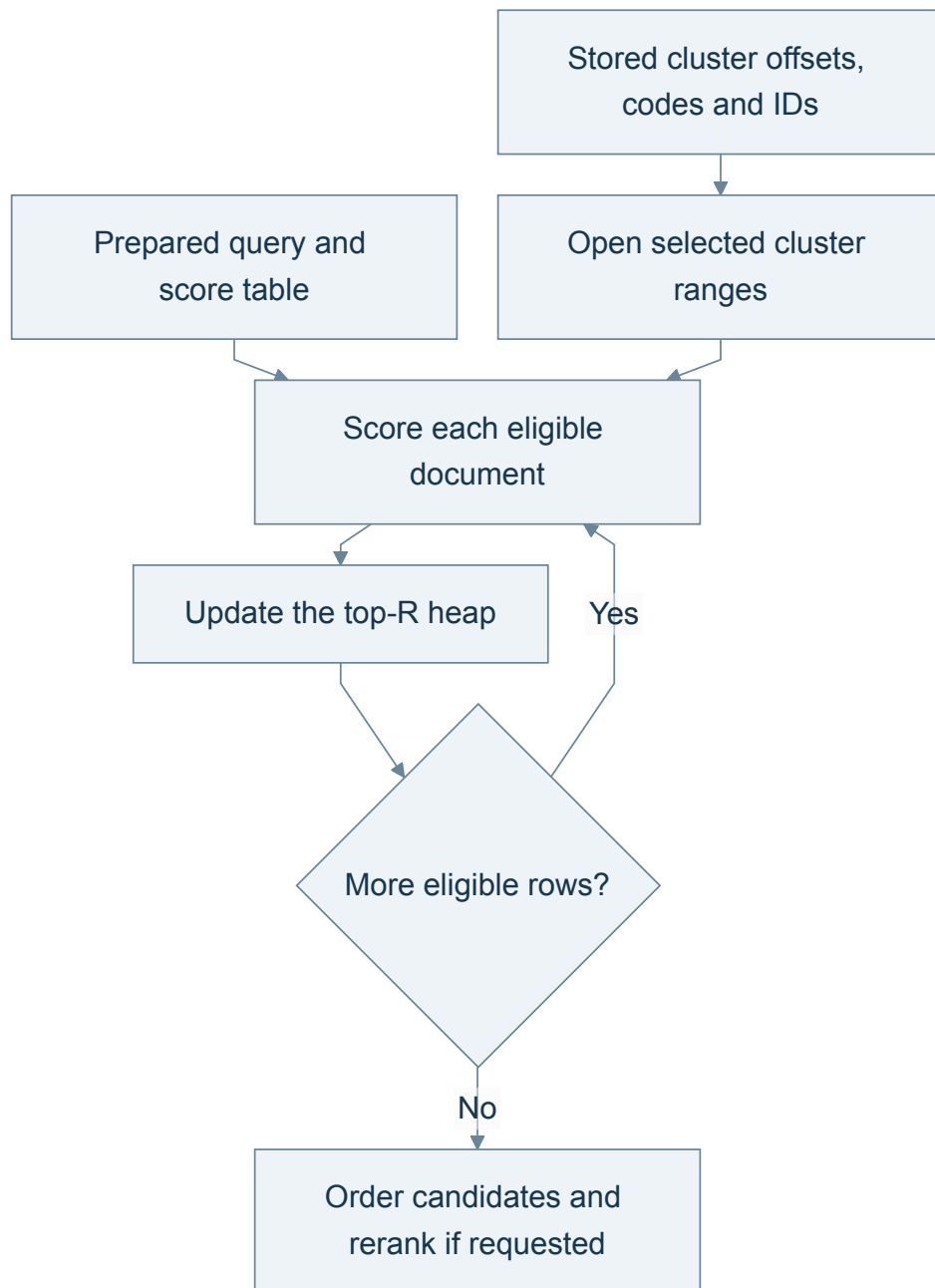


Figure 2: Approach A. The selected pool is smaller than the corpus only because of routing or filters.

3.1 Data layout

The common index from Equation (3) is sufficient. Cluster j occupies the row range between offsets j and $j + 1$. A row contains its packed code, and the corresponding ID array maps the row back to the original document.

Listing 5: Candidate search for Approach A.

```
ScanSelectedClusters(q, selected_clusters, eligible, table, R):
    best = empty top-R heap
    for cluster j in selected_clusters:
        for row in eligible rows of j:
            score = ScoreDocument(code[row], table)
            best.offer(score, document_id[row])
    return best in descending score, breaking ties by ID
```

The loop is over N_{elig} rows, not necessarily NP/C . If all clusters are searched and no filter removes documents, $N_{\text{elig}} = N$. Empty filtered clusters require no document scoring, although finding that they are empty can still have a cost.

3.2 Work count

Step	Work	Live temporary data
Locate cluster ranges	P offset/range accesses	selected cluster IDs
Visit eligible rows	N_{elig} row iterations	current row/iterator
Score rows	$N_{\text{elig}}G$ lookup terms	shared score table, accumulator
Keep best R	$H_s(N_{\text{elig}}, R)$ selection operations	top- R heap
Return candidates	sort at most R results if needed	result array or heap storage

Let c_{range} denote range setup and $T_{\text{rank}}(R)$ denote final candidate ordering. Then

$$T_{A,\text{candidates}} = Pc_{\text{range}} + N_{\text{elig}}Gc_{\text{lut}} + H_s(N_{\text{elig}}, R)c_{\text{heap}} + T_{\text{rank}}(R). \quad (14)$$

The general end-to-end latency is

$$T_A = T_{\text{common}} + T_{A,\text{candidates}} + T_{\text{finish}}(R'). \quad (15)$$

For the stated heap implementation, the candidate-stage time complexity is

$$O(P + N_{\text{elig}} \lceil d/g \rceil + N_{\text{elig}} \log(R + 1) + R \log(R + 1)).$$

The lookup term count is not a claim that each term is exactly one CPU instruction.

3.3 Data reads and memory

The number of packed-code data bytes read for scoring is

$$V_{A,\text{codes}} = N_{\text{elig}} s_{\text{code}}. \quad (16)$$

ID reads, filter data, table accesses and result-heap accesses are additional traffic. A cache may satisfy many of them. This formula therefore counts only code data; it does not measure all reads from main memory. Sequential and scattered rows may have different effective c_{lut} or c_{row} .

Index storage and temporary memory during candidate search are

$$M_{A,\text{index}} = M_{\text{base}}, \quad (17)$$

$$Q_{A,\text{search}} = Q_{\text{base}} + Q_{\text{row iterator}} + Q_{\text{scan workspace}}. \quad (18)$$

A streaming scan does not need an N -element score array. A matrix-based batch implementation may choose to allocate such an array; then that allocation must be added. Two implementations can therefore return the same ranking while using different amounts of peak RAM.

3.4 Routing example

The eight documents from Section 1 are partitioned into

$$\mathcal{C}_0 = \{1, 3, 4, 6\}, \quad \mathcal{C}_1 = \{2, 5, 7, 8\}.$$

This is an illustrative fixed partition, not a claim about the output of a particular clustering run. Suppose the router opens only $\mathcal{C}_1$. A computes scores 1.2, 0.6, 0.8, -0.2 and returns documents 2 and 7. The global top two are 1 and 2, so recall is $1/2$. The scan is exact within the opened cluster; document 1 is excluded by routing.

Opening both clusters returns documents 1 and 2. A scan can therefore be exact within its selected pool while its global recall remains below one.

3.5 SIMD

SIMD means one instruction operates on several values at once. It can accelerate code extraction, lookup accumulation, and selection. It does not by itself change the number of stored bits. There are two useful ways to model its effect.

Calibrated model. The scoring routine's time per work unit, c_{lut} , is measured for the relevant dimensions and access pattern. If that rate already includes selection, a combined $T_{\text{score+select}}$ excludes a separate heap-cost term to avoid double counting.

Resource model. If a scoring routine performs U_{op} compute units and causes V_{DRAM} bytes of DRAM traffic, compute throughput Γ and memory bandwidth β give the lower bound

$$T_{\text{kernel}} \geq \max\{U_{\text{op}}/\Gamma, V_{\text{DRAM}}/\beta\}. \quad (19)$$

An effective SIMD factor s_{simd} can change Γ to $s_{\text{simd}}\Gamma_0$. It cannot make the bandwidth term disappear. Random access, synchronization and fixed overhead can put actual time above the bound. This resource estimate is not added to a fitted coefficient that already includes the same work.

For multiple queries, a float score matrix can be written as $QB^\top$, where rows of B are expanded binary codes. Matrix operations can compute this product, but code expansion, batch formation, device transfers and score-buffer storage incur costs. Fast GPU batch processing does not necessarily imply low single-query latency.

Accuracy

For a fixed eligible pool, A returns its exact binary top R under the stated tie rule. With every eligible cluster searched, it has 100% binary recall. With partial routing, its global binary recall is limited by which true neighbors reach the selected pool. The scan cannot restore documents missed by routing, quantization or an earlier filter.

4 Bitplane search

Approach B keeps the same document score and common routing stage. Its additional index stores the document bits by coordinate instead of only by row. This layout permits testing one sign across many document positions at once.

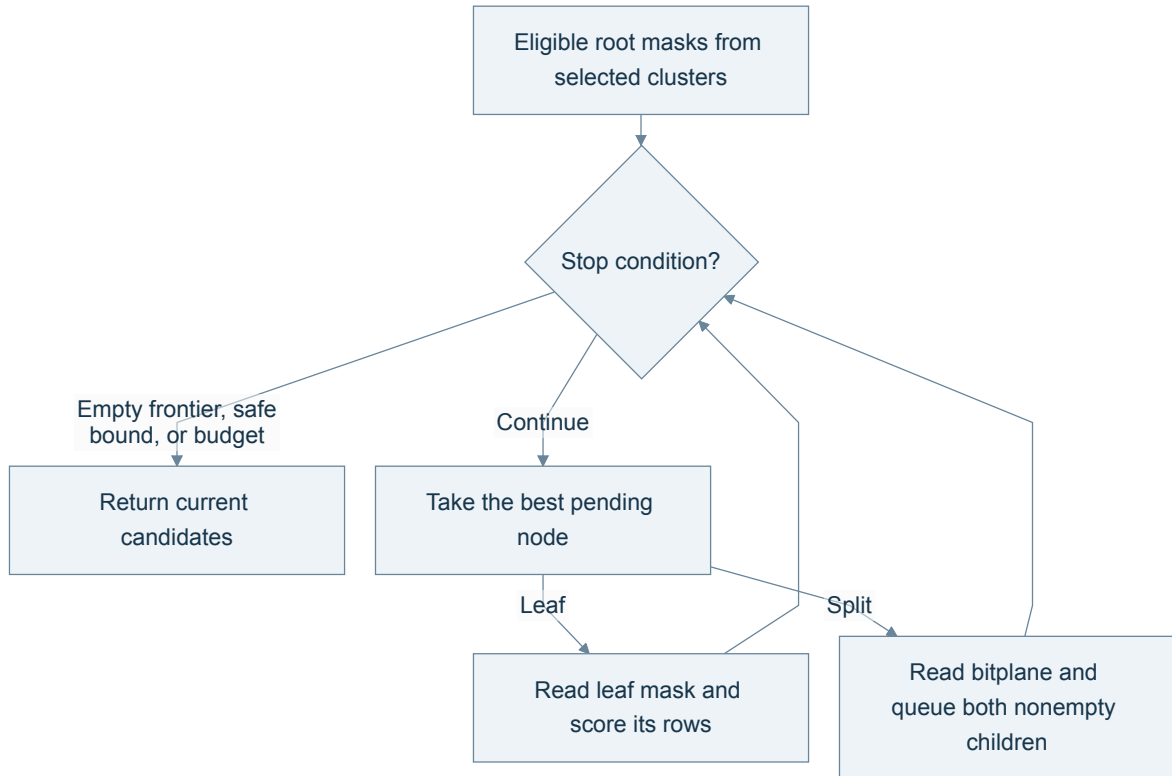


Figure 3: Approach B. A pending node has a set of active documents and an optimistic score bound.

4.1 Bitplane layout

For the opened example cluster $\mathcal{C}_1 = (2, 5, 7, 8)$, the document codes are 1011, 1110, 1001, 0011. Writing one row per coordinate gives

	doc 2	doc 5	doc 7	doc 8
bitplane 1	1	1	1	0
bitplane 2	0	1	0	0
bitplane 3	1	1	0	1
bitplane 4	1	0	1	1

Each row is a bitmap over document positions, not a list of full document vectors. An active bitmap identifies the document positions that still belong to the current node. Requiring coordinate i to be 1 corresponds to intersecting the active bitmap with plane i . Requiring 0 uses its complement, with unused padding positions masked out.

Listing 6: Build the additional bitplane index.

```
BuildBitplanes(cluster_codes, d, w):
    for each cluster j:
        W[j] = ceil(number_of_rows(j) / w)
        planes[j] = d arrays of W[j] zero words
        for local row u in j:
            for coordinate i in 0 .. d-1:
                if code[j, u, i] is 1:
                    set bit (u mod w) in planes[j, i, floor(u/w)]
    return planes
```

For this layout, the extra index storage is exactly

$$M_{\text{planes}} = db_w \sum_{j=1}^C W_j, \quad M_{B,\text{index}} = M_{\text{base}} + M_{\text{planes}}. \quad (20)$$

Without per-cluster padding this is $Nd/8$, as many data bytes as a second copy of the binary codes. For example, ten billion 256-bit codes occupy 320 GB; their bitplanes add approximately another 320 GB, not 32 TB. Padding and other index structures are additional.

4.2 Branch order

At coordinate i , choosing the query's sign adds no penalty; choosing the opposite sign adds $a_i = |q_i|$. Sorting coordinates by descending a_i tests the largest potential penalties first. For the four-bit query, the order is 1, 2, 3, 4 with weights 0.7, 0.4, 0.2, 0.1.

In $\mathcal{C}_1$, matching bit 1 keeps documents 2, 5, 7; the alternative contains document 8 with accumulated penalty 0.7. Matching bit 2 next keeps 2, 7; document 5 moves to another pending node with penalty 0.4. This order prioritizes larger penalties, but does not by itself justify discarding either alternative.

Example: A greedy path can miss the best document

Consider $q = (0.6, 0.5, 0.4, 0.3)$ and two documents: 0111 and 1000. The first has penalty 0.6 and score 0.6. The second has penalty 1.2 and score -0.6 . A greedy choice of the matching first bit selects 1000 and misses the better document. The largest individual weight is smaller than the sum of several remaining weights.

4.3 Score bounds

Let a node have accumulated penalty p on coordinates already tested. Untested coordinates cannot contribute a negative mismatch penalty. Consequently,

$$U(\text{node}) = A - 2p \tag{21}$$

is an upper bound: no active document can score higher than this best possible score. If the result heap is full and U is strictly below its worst retained score, that node cannot improve the result. For equal scores, search continues unless the ID tie rule also excludes every document in the node.

The **frontier** is the collection of pending nodes. A best-first search takes the node with the largest upper bound, or equivalently the smallest penalty. This best possible score may use a sign combination that no actual document has. A bound far above the actual scores can retain many pending nodes that never improve the result.

Listing 7: Dense bitplane search across the selected clusters.

```

SearchBitplanes(q, selected_clusters, eligible, table, R, L, B_node):
    order = coordinates sorted by decreasing abs(q[i])
    frontier = empty min-penalty priority queue
    best = empty top-R heap
    for each selected cluster j:
        active = bitmap of eligible rows, with physical width W[j]
        if active is not empty:
            frontier.push(j, depth=0, penalty=0, active)
    visits = 0
    while frontier is not empty:
        if best is full and best_possible(frontier) < best.worst_score:
            break
        if a finite B_node is set and visits == B_node:
            break
        node = frontier.pop_smallest_penalty()
        visits += 1
        if node.count <= L or node.depth == d:
            for word in node.active:                # includes zero words
                for each set bit in word:
                    score its document and offer it to best
            continue
        i = order[node.depth]
        matching = active AND preferred_sign_plane(j, i, q)
        opposite = active AND NOT preferred_sign_plane(j, i, q)
        queue nonempty matching child with the same penalty
        queue nonempty opposite child with penalty + abs(q[i])
    return best

```

L is the leaf-size threshold and B_{node} the total node budget across all probes. With no budget limit, score bounds permit exact stopping; a finite budget can stop before the remaining nodes are safely ruled out. A random exploration probability changes which pending node is visited next. It is not a direct formula for recall or a synonym for the number of nodes.

4.4 Work count

For each selected cluster, let

$$s_j = \text{visited split nodes}, \quad \ell_j = \text{visited leaf nodes}, \quad f_j = \text{documents fully scored}.$$

Define $V = \sum_j (s_j + \ell_j)$ and $F = \sum_j f_j$. For a disjoint traversal without rescoring, $F \leq N_{\text{elig}}$. A finite node budget gives $V \leq B_{\text{node}}$, but does not determine F . The two main bitmap counts are

$$U_{\text{split}} = \sum_j s_j W_j, \quad U_{\text{leaf}} = \sum_j \ell_j W_j. \quad (22)$$

A split word in the reference implementation reads an active word and a plane word, produces two child words, and counts matches. It includes two ANDs, a population count, complements and an accumulation. Allocating zero-filled child arrays can add two initialization writes. A leaf still scans all words in its active bitmap before visiting its set bits.

With separate effective coefficients,

$$T_{B,\text{candidates}} = T_{\text{order}} + T_{\text{roots}} + U_{\text{split}}c_{\text{split}} + U_{\text{leaf}}c_{\text{leafword}} + Vc_{\text{node}} \\ + FGc_{\text{lut,leaf}} + H_s(F, R)c_{\text{heap}} + T_{\text{rank}}(R), \quad (23)$$

$$T_B = T_{\text{common}} + T_{B,\text{candidates}} + T_{\text{finish}}(R'). \quad (24)$$

Here $T_{\text{order}} = O(d \log d)$ and $T_{\text{roots}} = O(\sum_j W_j)$ for dense root construction. c_{node} may be replaced by explicit queue-comparison, mask-allocation and copy terms. The accounting convention must be consistent to avoid counting an allocation twice.

4.5 Full-width bitmap cost

Example: Few scores do not imply little work

In the saved one-million-document run, $w = 64$ gives $W = 15,625$. A node budget of 32,768 produced a mean 22,557.88 split visits and 10,210.12 leaf visits. Therefore

$$U_{\text{split}} = 352,466,875, \quad U_{\text{leaf}} = 159,533,125.$$

Their sum is 512 million word-loop visits. About 163,766 documents were fully scored. With $g = 4$, that is 10.48 million score-table terms, versus 64 million terms for the scan. The reduction in scoring was accompanied by hundreds of millions of bitmap visits. Word visits and score-table terms are not equal-cost instructions.

If a tree were fully explored, perfectly balanced, and every internal node had two nonempty children, then with L documents per leaf there would be roughly n_j/L leaves and one fewer split. This gives a dense bitmap term of order

$$O\left(\sum_j \frac{n_j^2}{Lw}\right). \quad (25)$$

These simplifying assumptions explain the cost; they do not predict it for real embeddings. Fixed-bit splits can have only one nonempty child. A finite traversal also leaves unvisited nodes. Thus the actual split count need not be half the visited-node count.

A single greedy path visits only about $\log_2(n_j/L)$ levels in a balanced example, but still incurs a full W_j at each level and has no general recall guarantee. In the broad worst case, a fixed-coordinate trie can contain $O(n_j d)$ nonempty prefix nodes, making dense traversal work as large as $O(d \sum_j n_j^2/w)$ before scoring and queue costs. Growing N alone does not guarantee that branching will become faster than a scan.

4.6 Pending masks and memory

Let $\mathcal{F}(t)$ be the frontier at time t , and let $j(v)$ identify a node's cluster. If a node record needs b_{node} bytes and bitmap capacity is fully allocated,

$$Q_{B,\text{frontier}}(t) = \sum_{v \in \mathcal{F}(t)} (b_{\text{node}} + b_w W_{j(v)}). \quad (26)$$

Capacity growth and allocator overhead can be represented by a factor $\gamma_B \geq 1$ or, preferably, by measured capacities. During a split, the parent and newly allocated children can be in memory together. An upper estimate for candidate-stage memory is

$$Q_{B,\text{search}} = Q_{\text{base}} + Q_{\text{order}} + \max_t [\gamma_B Q_{B,\text{frontier}}(t) + Q_{\text{current}}(t) + Q_{\text{children}}(t)]. \quad (27)$$

For nonempty, disjoint pending sets, frontier count is at most N_{elig} . Starting with at most P roots, each processed binary split can increase that count by at most one, so a finite budget also gives $|\mathcal{F}| \leq P + B_{\text{node}}$. Multiplying that bound by the largest possible mask gives an upper estimate that can be much too high. It should not be reported as measured RAM.

When branching helps

B can reduce latency when the time saved by avoiding full scoring exceeds the added bitmap, queue and allocation time. Favorable conditions include large early query penalties, small physical bitmaps, score bounds close to the actual scores, and few pending nodes. Unlimited search is exact inside the selected pool under valid bounds and ties. Recall for budget-limited search requires measurement; its budget is not itself an accuracy percentage.

5 Key lookup

Approach C retrieves documents through a key directory rather than repeated document-mask intersections. The search begins with the signs preferred by the query and considers alternative keys in increasing mismatch penalty.

This method still starts with the same selected clusters. A global hash table is the special case $C = P = 1$. Clustered and global formulas are kept distinct.

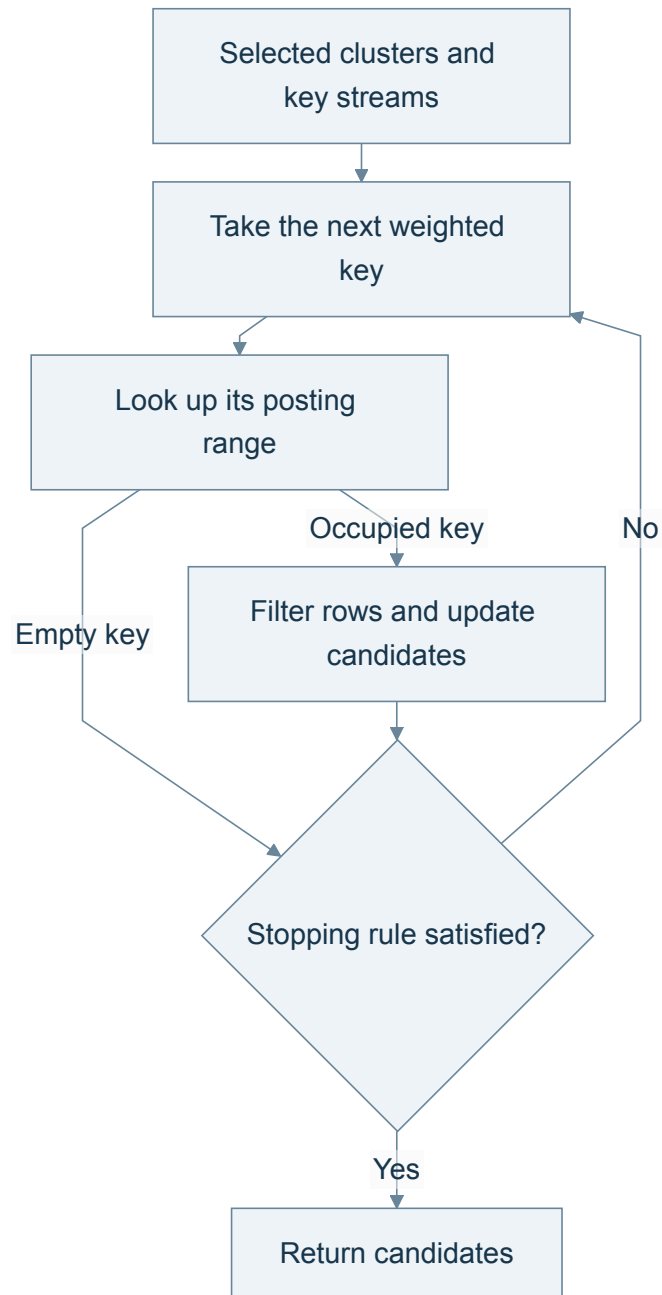


Figure 4: Approach C. Each cluster has a key directory and postings. Empty lookups still consume work.

5.1 Full and short keys

Let I_h be the fixed set of h lookup coordinates, often the first h positions. A **full key** has $h = d$. Its weighted penalty determines the complete binary score. A **short key** has $h < d$. It is used to rank a prefix; the full d -bit score then ranks the retrieved candidates.

A full-key search can be exact if it enumerates keys in true query-weighted order and finishes checking all ties at the result cutoff. A short-key candidate quota does not have that guarantee. It may still provide useful candidates, and a valid full-score stopping bound can make it exact. These guarantees depend on the stopping rule rather than the term “walk backwards.”

5.2 Key directory

For each cluster, a directory maps a key to its postings: the local row references of documents with that key, stored as one range. There may be many documents per key. Finding a directory entry without finding its document rows is not enough to produce results.

Listing 8: Build a clustered key index.

```
BuildKeyIndex(cluster_codes, lookup_coordinates):
    for each cluster j:
        groups = group local row references by their lookup key
        concatenate the groups into a posting array
        for each occupied key:
            directory[j][key] = (posting_start, posting_count)
    return directories, postings
```

A **dense directory** has an offset for every possible key. A **sparse hash table** only stores keys that have documents, but needs space for keys, slots, and unused capacity. A sorted-key array is another option: it can use less overhead, but a lookup uses comparisons or another search structure instead of an assumed constant-time hash access.

5.3 Fixed routing bits

If sign-prefix routing fixes some lookup coordinates inside cluster j , those coordinates cannot vary within it. Let r_j be the number of lookup coordinates fixed by that cluster, and let

$$h_j = h - r_j, \quad U_j \leq \min(n_j, 2^{h_j}). \quad (28)$$

U_j counts the keys that actually occur on the coordinates still free to vary. For first- r -bit routing and a first- h -bit lookup with $h \geq r$, $h_j = h - r$. The fixed coordinates add a *constant* penalty for that cluster; they do not create 2^r additional keys inside it. Ordinary k-means clusters do not generally fix any sign coordinate, so this simplification does not follow from their cluster ID.

Example: Two fixed bits and a two-bit local key

Suppose a four-bit code is routed by its first two bits. Inside cluster 10, codes 1010, 1011, 1000, 1001 need only the local keys 10, 11, 00, 01. A lookup for the full prefix must include the cluster’s constant routing penalty when it is compared with a different cluster’s stream. The number of local possibilities is four, not sixteen.

Both child costs are no smaller than the parent cost. Every nonempty subset has one parent: if the coordinate before its largest coordinate is also present, remove the largest one; otherwise replace the largest coordinate by its predecessor. The construction therefore visits every subset without a duplicate-state hash set.

Each queued state must store enough information to construct the key. A full bitmask representation costs $b_w \lceil h_j/w \rceil$ bytes per state, plus its penalty, last coordinate and queue metadata. Copying that mask into children is work. A persistent representation can change this cost, but it then needs its own traversal and memory accounting.

Example: Full-key example

For $q = (0.6, 0.5, 0.4, 0.3)$, the ideal code is 1111. The first penalties include 0, 0.3, 0.4, 0.5, 0.6, 0.7, and so on. The code 0111 is found by flipping coordinate 1 at penalty 0.6. The code 1000 has penalty 1.2 and comes later. Complete weighted enumeration finds the better document first, even though it disagrees on the query's largest coordinate. Several empty keys may be checked before this document is found.

5.6 Lookup and stopping

For short keys, a simple implementation allocates candidate targets m_j to probed clusters. A more flexible implementation merges their next-prefix-key scores and draws candidates from the best available stream. Whichever policy is used must be a parameter of the experiment; a per-cluster target and a global target are not interchangeable.

Listing 10: Short-key candidate search with explicit per-cluster targets.

```
SearchShortKeys(q, selected_clusters, h, targets m[j], R):
    candidates = empty collection
    for each selected cluster j:
        stream = SortedFlipSets(free lookup weights in j)
        found = 0
        for (flip_set, variable_penalty) in stream:
            key = ideal_free_key(q, j) XOR flip_set
            rows = directory[j].lookup(key)      # may be empty
            keep eligible rows from this posting range
            found += number of kept rows
            append those rows to candidates
            if found >= m[j] and final prefix-penalty ties are finished:
                break
    score every candidate with the full d-bit score
    keep the best R with the same score/ID rule as A
    return best R
```

A full-key exact version instead merges the cluster streams by their *total full-key* penalty, including fixed-sign penalties. After at least R documents have been found, enumeration continues through every key that can tie or exceed the R th document. This produces the exact binary top R within the selected pool. If unprobed clusters remain, global exactness additionally requires their score bounds to be ruled out or those clusters to be searched.

A short key can also stop exactly. Let $T_h(b)$ be the prefix score and $A_{\text{tail}} = \sum_{i \notin I_h} |q_i|$. Every

document in an unvisited prefix key satisfies

$$S(q, b) \leq T_h(b) + A_{\text{tail}}. \quad (31)$$

If the best possible score of any unvisited prefix is strictly below the current worst top- R score, the result is exact. Otherwise, exact search must continue even when a nominal candidate quota has been reached. A larger tail allows a higher best possible score, so this test rules out fewer prefixes. Fixed routing-coordinate contributions must be included consistently; no coordinate may be counted twice.

5.7 Reusing key scores

Document rescoreing is not always necessary. With a full key and the true query weights, its complete score is already $A - 2\pi_q(\text{key})$. Every document in that key’s posting range has that same binary score. The search can add their IDs to the best- R heap without unpacking and rescoreing each code. For a short key, the known prefix contribution can similarly be reused while evaluating only the remaining coordinates.

To make that implementation choice visible, let $G_{C,j}$ be the number of score-table groups actually evaluated per candidate in cluster j . The simple pseudocode above has $G_{C,j} = G$. A full-key implementation reusing its exact key score can have $G_{C,j} = 0$. A tail-only scorer can have approximately $\lceil d_{\text{remaining}}/g \rceil$ groups, with alignment and fixed routing coordinates handled explicitly. It still incurs the cost of postings, key-score arithmetic, selection, and any later float reranking. Reusing the score saves work without reducing recall.

The reference storage formula retains the common packed rows. A layout designed for full keys can also avoid a redundant per-document code copy by using its stored full keys, provided the revised layout accounts for the storage of IDs, postings, and keys. Equation (30) is one explicit layout, not a universal lower bound on every possible key index.

5.8 Prefix relaxation

Search can also proceed by removing constraints. This variant removes a bit constraint instead of flipping a bit and probing one additional exact key. For example, **101*** means either **1010** or **1011**. That is a set of keys. A plain exact-key hash table does not answer this wildcard request in one ordinary lookup.

A fixed-order **prefix trie** stores paths by successive key coordinates. A node identifies an occupied prefix. If document rows are ordered by key within a cluster, each node can store the start and end of a contiguous posting range. Moving to its parent relaxes the last fixed coordinate. It does not require copying every descendant’s posting list into every ancestor.

Example: Relaxing the key inside the example cluster

For $\mathcal{C}_1 = \{2, 5, 7, 8\}$ and ideal key 1010:

Constraint	Matching documents	Newly visited documents
1010	none	none
101*	2	2
10**	2, 7	7
1***	2, 5, 7	5
****	2, 5, 7, 8	8

The parent range contains the previous range. Only the newly added sibling ranges need to be scanned if the implementation retains their boundaries. Otherwise repeated scans must be counted again. In this particular query the last coordinate is the cheapest one to relax. That is not true for every query.

Listing 11: A fixed-prefix relaxation variant of Approach C.

```

SearchByRelaxingPrefix(q, query_key, trie, table, R, candidate_target):
    path = follow query_key as far as the stored trie permits
    previous_range = empty
    best = empty top-R heap
    found = 0
    for node on that path, from deepest node toward the root:
        new_ranges = node.posting_range minus previous_range
        for eligible row in new_ranges:
            score its code and offer it to best
            found += 1
        previous_range = node.posting_range
    if found >= candidate_target:
        return best                                # approximate stopping rule
    return best

```

An exact version may stop early only after bounding *every* unvisited region against the current full-score heap. Finding enough IDs in a relaxed prefix does not prove this. The remaining score can still change the ordering.

Let Z_j be the number of stored trie nodes and b_{trie} their record size, including child links or equivalent navigation and range offsets. A simple uncompressed trie satisfies

$$Z_j \leq 1 + \sum_{u=1}^{h_j} \min(n_j, 2^u).$$

Its directory term replaces the hash-directory term by $\sum_j Z_j b_{\text{trie}}$. Path compression can reduce the actual node count. With range references, document postings are stored once; copying a posting list into every prefix node would instead add repeated references and can cost $O(Nh)$ reference storage.

For u_j ancestor/range visits and F_j total row visits, a reference cost is

$$T_{C,\text{trie}} = T_{\text{keyprep}} + \sum_j [h_j c_{\text{path}} + u_j c_{\text{range}} + F_j c_{\text{post}} + F_j G_{C,j} c_{\text{lut,gather}}] + T_{\text{select/rank}}. \quad (32)$$

A saved navigation path uses $O(h_j)$ records per active stream. The memory account also includes the candidate heap, gather buffers, and common query RAM. Here F_j must count repeated rows if they are revisited. The same common pipeline and final rerank terms are added for end-to-end latency.

A fixed trie can remove its last prefix coordinate at low cost. Dropping an arbitrary low-magnitude coordinate chosen for the query can produce several separate ranges, so it does not have the same constant-cost parent operation. A query-adaptive bit order requires additional traversal or a different index. This is the main difference between a prebuilt fixed trie and the query-ordered bitplane traversal.

5.9 Work count

Let H_j be the number of keys attempted in cluster j , including empty keys. Let F_j be the number of eligible posting rows retrieved and scored. Let E_j be the number of sorted-subset heap comparisons, C_j^{copy} the number of key-word copies, and let Q_j^{max} be the largest enumeration-queue size. Define $F = \sum_j F_j$.

For the two-child stream, $E_j = O(H_j \log(Q_j^{\text{max}} + 1))$ comparison work, and $C_j^{\text{copy}} = O(H_j \lceil h_j/w \rceil)$ for full-mask copies. Hashing a key also depends on its length and table layout. The time $c_{\text{lookup}}(h_j, \alpha)$ should cover the work for each emitted key outside heap comparisons and child-mask copies: key construction, fixed state bookkeeping, the directory lookup, and obtaining the current key score or bound. This time includes cache effects and hash collisions; a lookup is not one CPU operation.

$$T_{C,\text{candidates}} = T_{\text{keyprep}} + \sum_{j \in \mathcal{P}} \left[E_j c_{\text{enum}} + C_j^{\text{copy}} c_{\text{copy}} + H_j c_{\text{lookup}}(h_j, \alpha) + F_j c_{\text{post}} \right] \\ + \sum_j F_j G_{C,j} c_{\text{lut,gather}} + H_s(F, R) c_{\text{heap}} + T_{\text{rank}}(R), \quad (33)$$

$$T_C = T_{\text{common}} + T_{C,\text{candidates}} + T_{\text{finish}}(R'). \quad (34)$$

T_{keyprep} includes sorting lookup weights, initializing streams and constructing query keys. For P different free-coordinate sets, a straightforward upper count is $O(\sum_j h_j \log(h_j + 1))$. A global merge adds its own P -stream selection work, typically $O((\sum_j H_j) \log(P + 1))$, which must be included when that variant is used.

The hash directory can be built before the query, but query-specific weighted order depends on q . Precomputing a hash table does not precompute this order for every float query.

5.10 Query memory

Let $b_{\text{enum},j}$ include a key mask, penalty and queue metadata. For simultaneously active streams, the temporary memory for candidate search is

$$Q_{C,\text{search}} = Q_{\text{base}} + \max_t \left[\sum_j \gamma_C Q_j(t) b_{\text{enum},j} + Q_{\text{stream merge}}(t) + Q_{\text{posting buffer}}(t) + Q_{\text{gather}}(t) \right]. \quad (35)$$

A sequential per-cluster implementation can reuse enumeration memory, replacing the sum of live stream sizes by the maximum live cluster stream. A global full-score stream may need all selected streams active. These are different execution choices.

Postings can be streamed into a top- R heap; storing all F retrieved IDs is not mandatory. Storing them for batch reranking adds Fb_{ref} or the memory required for the actual bounded batch size. A small sparse dictionary does not guarantee a small queue while the query lists keys. With the two-child stream, the queue can grow by at most one per emitted state, but H_j itself can be very large.

Accuracy

Full-key enumeration in true penalty order can have 100% binary recall. Its limiting cost can be time or queue memory, especially when most keys are empty. Short-key search is a different tradeoff: it can collect useful candidates at low cost, but a fixed quota is not a recall guarantee. It becomes exact if a valid full-score bound proves that no unseen candidate can improve the result, or if it eventually scores every eligible document.

6 Recall analysis

A search that returns 80 candidates from a database of 256 reduces the remaining scoring work to those candidates. Recall depends on how many of the nearest documents are among those 80. This section derives both quantities under a small random-bit model, then extends the analysis to query weights and clusters.

6.1 Recall definition

Let $\mathcal{E}(q)$ be the eligible document set after the intended filter, across the whole database. Let $\mathcal{G}_K(q)$ be its exact top K under the chosen reference score, and let $\mathcal{R}_K(q)$ be the returned results. Then

$$\text{Recall@ } K(q) = \frac{|\mathcal{G}_K(q) \cap \mathcal{R}_K(q)|}{K}. \quad (36)$$

Both sets use the same tie rule, and mean recall averages this quantity over queries. Repeated timing measurements of one query do not provide independent quality observations.

If a candidate set $\mathcal{D}(q)$ is reranked with the *same exact reference score*, every true top- K document in that set survives into the final top K . Therefore

$$\text{Recall@ } K(q) = \frac{|\mathcal{G}_K(q) \cap \mathcal{D}(q)|}{K}. \quad (37)$$

A different reranking score breaks that identity. In particular, exact binary candidates may still miss the true float-vector top K or the documents relevant to a human question.

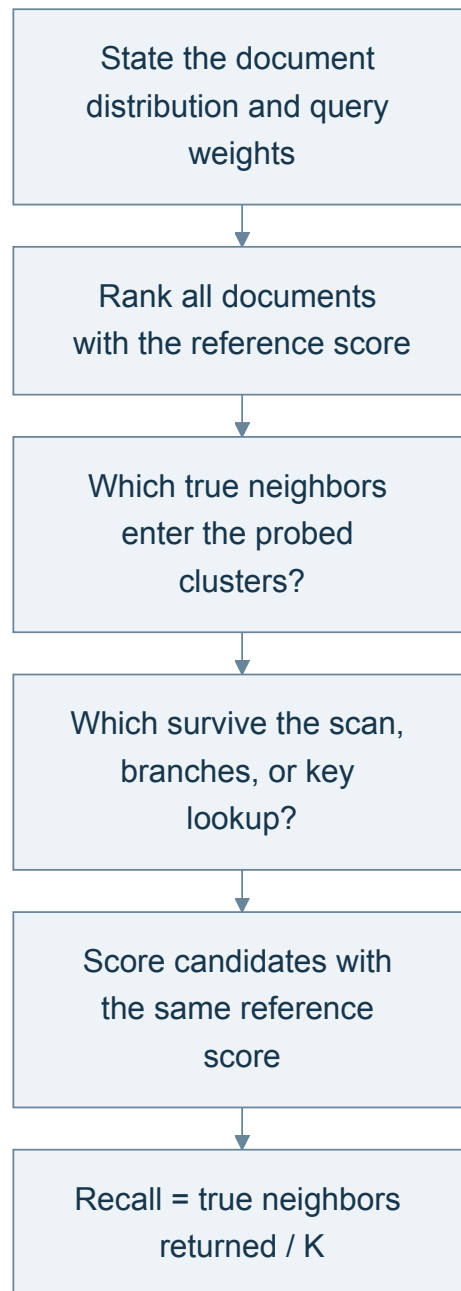


Figure 5: Follow the same true neighbors through every stage. The denominator stays K throughout the recall calculation.

6.2 A small random-bit model

Suppose a document has d bits. We sample N documents independently, and each bit is independently zero or one with probability $1/2$. Duplicate codes are allowed: two different documents can have the same bits. Their row IDs resolve score ties, and those IDs are assigned independently of the sampled codes.

Initially, every query coordinate has the same magnitude. A document's score therefore depends only on how many bits disagree with the query. We can write the query's desired pattern as all zeros without changing any probabilities. The number of mismatches is called the *Hamming distance*. A smaller distance means a better score.

The search checks h fixed coordinates and accepts a document if at most t of them disagree. It then scores every accepted document using all d bits. For example, with $h = 4$ and $t = 1$, it opens the keys 0000, 0001, 0010, 0100, and 1000. This is a precise candidate rule. It does not stop halfway through a key's documents when a quota is reached.

These assumptions define an illustrative model. Equal weights permit direct counting, and independent bits make every pattern equally likely. Later subsections relax these assumptions. There are no clusters or eligibility filters in this first calculation.

6.3 Nearest-document distance

Let Z be one random document's distance from the query. To have exactly m errors, choose which m of the d positions are wrong. There are $\binom{d}{m}$ choices, out of 2^d equally likely codes. Thus

$$\Pr(Z = m) = \frac{\binom{d}{m}}{2^d}, \quad F_d(m) = \Pr(Z \leq m) = \frac{1}{2^d} \sum_{j=0}^m \binom{d}{j}. \quad (38)$$

This is the binomial distribution with parameters d and $1/2$. The function F_d sums the probabilities of all distances through m . This accumulated probability defines a cumulative distribution function, or CDF. Define $F_d(-1) = 0$.

Let M be the distance of the nearest document among all N . If one document has probability $1 - F_d(m)$ of being farther than m , independence gives probability $[1 - F_d(m)]^N$ that *all* documents are farther away. Subtracting from one gives

$$\Pr(M \leq m) = 1 - [1 - F_d(m)]^N. \quad (39)$$

The probability of exactly distance m follows by subtracting the probability of a smaller distance:

$$\Pr(M = m) = [1 - F_d(m-1)]^N - [1 - F_d(m)]^N. \quad (40)$$

A random document usually has many mismatched bits, whereas the nearest of hundreds of documents has unusually few. Recall therefore depends on the distribution of the nearest distance, rather than the distance of an arbitrary document.

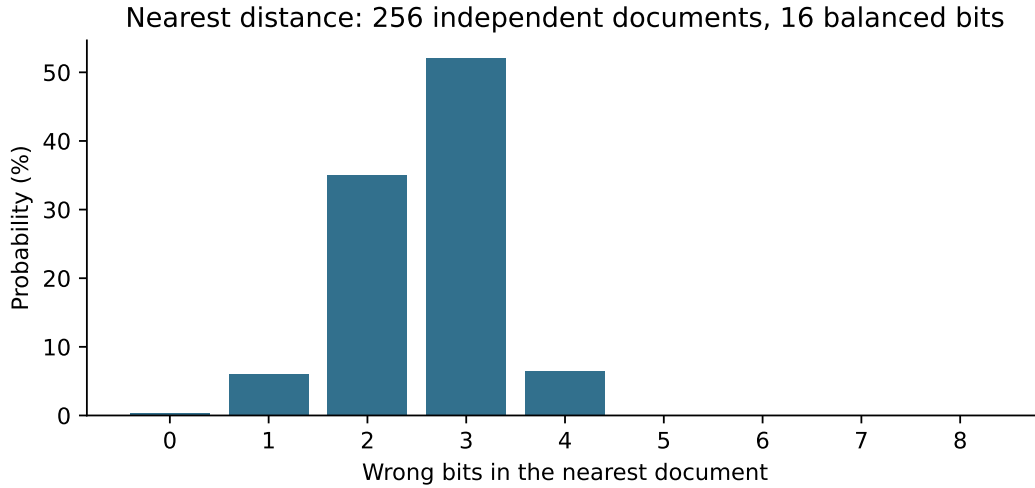


Figure 6: With 256 random 16-bit documents, the nearest distance is usually two or three. Assuming it is always two changes the recall calculation.

6.4 Key acceptance probability

Suppose the nearest document has exactly m wrong bits. Under the uniform model, every set of m error positions is equally likely. Let A count how many of those errors fall in the h lookup positions. To have exactly a lookup errors, choose a positions from the lookup bits and $m - a$ from the other bits. Divide by all possible placements of the m errors:

$$\Pr(A = a \mid M = m) = \frac{\binom{h}{a} \binom{d-h}{m-a}}{\binom{d}{m}}. \quad (41)$$

This is a hypergeometric distribution. It counts positions drawn from a fixed number of errors. The error count is already fixed here, so treating these positions as independent trials would change the model. A binomial coefficient is zero when its lower argument is negative or larger than its upper argument.

The search accepts at most t lookup errors. Summing the allowed placements gives:

$$g_{h,t}(m) = \Pr(A \leq t \mid M = m) = \frac{\sum_{a=0}^{\min(h,t)} \binom{h}{a} \binom{d-h}{m-a}}{\binom{d}{m}}. \quad (42)$$

Even after we know the nearest distance, every placement of its errors is equally likely because the score uses only the total number of errors and the ID tie rule does not favor a particular pattern. This symmetry depends on the equal-weight assumption.

Example: Two wrong bits in a 16-bit document

Take $d = 16$, $h = 4$, and suppose $m = 2$. There are $\binom{16}{2} = 120$ ways to place the errors. If the first four bits must match perfectly, both errors must be in the other twelve bits: $\binom{12}{2} = 66$ placements survive. Recall, given two total errors, is $66/120 = 55\%$. Allow one error in the first four bits. Another $4 \times 12 = 48$ placements now survive, giving $(66 + 48)/120 = 95\%$. This 95% result is conditional on *two total errors*. It does not yet account for other possible distances of the nearest document.

6.5 Expected top-one recall

For top one, recall is one if the true nearest document is accepted and zero otherwise. Its expected value is therefore its probability of being accepted. For each possible nearest distance, multiply the probability of that distance by the probability of passing the prefix check. Add the results:

$$\mathbb{E}[\text{Recall@1}] = \sum_{m=0}^d \Pr(M = m) g_{h,t}(m). \quad (43)$$

For $N = 256$, $d = 16$, and $h = 4$, this produces the table below. The candidate counts are derived in the next subsection.

Allowed errors t	Keys opened	Expected documents	Expected Recall@1
0	1	16	46.40%
1	5	80	89.60%
2	11	176	99.45%
4	16	256	100.00%

For one allowed lookup error, the result is 89.60%, rather than the conditional 95% above. In this example, the nearest distance is two about 35.05% of the time and three about 52.05% of the time. A three-error neighbor is more likely to put too many errors in the lookup bits. Equation (43) includes both cases, along with every other possible distance.

Listing 12: Calculate expected top-one recall under the random-bit assumptions.

```
ExpectedRecall(N, d, h, allowed_errors):
    recall = 0
    for m in 0 .. d:
        nearest_probability = (1 - F_d(m-1))^N - (1 - F_d(m))^N
        passing_placements = 0
        for a in 0 .. min(h, allowed_errors):
            passing_placements += choose(h, a) * choose(d-h, m-a)
        survival = passing_placements / choose(d, m)
        recall += nearest_probability * survival
    return recall
```

6.6 Candidate and lookup counts

Opening all h -bit keys through radius t requires

$$H(h, t) = \sum_{a=0}^t \binom{h}{a}, \quad 0 \leq t \leq h. \quad (44)$$

Each random document has probability $p_{\text{pass}} = H(h, t)/2^h$ of belonging to one of those keys. If F counts accepted documents, then

$$F \sim \text{Binomial}(N, p_{\text{pass}}), \quad \mathbb{E}[F] = Np_{\text{pass}}, \quad \text{Var}(F) = Np_{\text{pass}}(1 - p_{\text{pass}}). \quad (45)$$

For $t = 1$, five of sixteen keys are opened. The expected candidate count is $256 \times 5/16 = 80$, with standard deviation $\sqrt{55} \approx 7.42$ documents. A particular corpus need not return exactly 80.

The 31.25% candidate fraction also differs from the 89.60% recall: nearest documents are more likely than arbitrary documents to pass this filter.

For C , H counts lookup attempts, including attempts that return nothing. The documents in those keys supply the posting and scoring work. A short-key query that scores all accepted rows might have expected time

$$\mathbb{E}[T_C] = T_{\text{prepare}} + Hc_{\text{lookup}} + \mathbb{E}[F](c_{\text{post}} + c_{\text{score}}) + \mathbb{E}[T_{\text{select}}],$$

with enumeration and routing added when the implementation performs them. Selection time can depend nonlinearly on the returned rows; substituting $\mathbb{E}[F]$ into every time term is another approximation. Use the full cost model in Equation (79) for the actual layout.

Storing every candidate ID at once takes Fb_{id} bytes. Streaming postings through a bounded top- K heap can avoid that particular full candidate array. Neither choice removes the index itself, and an expected candidate count is not a peak-RAM guarantee.

6.7 Expected top- K recall

For top K , the K selected rows can lie at different distances: some may have zero errors, some one, and others more. The calculation therefore requires the expected number of selected rows at each distance. Assume $1 \leq K \leq N$.

Let X_m count corpus documents whose distance is at most m . Since every document independently has probability $F_d(m)$ of meeting that condition,

$$X_m \sim \text{Binomial}(N, F_d(m)).$$

Exactly $\min(K, X_m)$ of the true top- K rows lie within that radius. There cannot be more than K , and when fewer than K documents qualify, every qualifying document belongs to the top K . Define

$$G_K(p) = \mathbb{E}[\min(K, X)], \quad X \sim \text{Binomial}(N, p). \quad (46)$$

Then the expected number of top- K rows at exactly distance m is

$$w_m = G_K(F_d(m)) - G_K(F_d(m-1)).$$

The same placement calculation applies to each row at that distance. Therefore

$$\mathbb{E}[\text{Recall@ } K] = \frac{1}{K} \sum_{m=0}^d w_m g_{h,t}(m). \quad (47)$$

This includes tied distances and duplicate codes under the stated ID rule. For $K = 1$, $G_1(p)$ is $1 - (1 - p)^N$, so it reduces to Equation (43). For $K = N$, every document is in the reference set and recall becomes the candidate fraction. Opening every key gives recall one for every K .

The appendix gives a numerical method for evaluating G_K . The formula accounts for the varying K -th distance without replacing it with a fixed estimated radius. It assumes a fixed prefix acceptance rule. Stopping after the first R postings is a different rule whose boundary depends on the sampled corpus.

6.8 Routing and local recall

Let $\mathcal{U}(q)$ be the eligible pool in the probed clusters. Define

$$r_{\text{route}}(q) = \frac{|\mathcal{G}_K(q) \cap \mathcal{U}(q)|}{K}, \quad r_{\text{local}}(q) = \frac{|\mathcal{G}_K(q) \cap \mathcal{D}(q)|}{|\mathcal{G}_K(q) \cap \mathcal{U}(q)|}.$$

When the denominator is positive and $\mathcal{D} \subseteq \mathcal{U}$,

$$\text{Recall@ } K(q) = r_{\text{route}}(q)r_{\text{local}}(q). \quad (48)$$

If no true top- K result enters the selected pool, total recall is zero. This multiplication is exact when both factors refer to the *same true global neighbors*. However, multiplying an average routing recall by an average local recall is generally incorrect:

$$\mathbb{E}[r_{\text{route}}r_{\text{local}}] = \mathbb{E}[r_{\text{route}}]\mathbb{E}[r_{\text{local}}] + \text{Cov}(r_{\text{route}}, r_{\text{local}}).$$

The covariance term records whether the two recall values tend to rise or fall together across queries. Recall against a cluster's local top K can also differ from recall for the particular global neighbors routed into it. Predictions for clustered search based on branch measurements from a single flat pool must preserve this distinction.

For A, local binary search is exact. For unlimited B with valid bounds, it is exact as well. For full-key C with complete weighted ordering, it is exact. Their global recall can still be limited by routing. Finite node budgets and short-key quotas add a second possible loss.

6.9 Equivalent candidate sets

The probability calculation describes a set of documents. Several algorithms can reach that same set. If they score those documents with the same full score and use the same tie rule, they have the same recall. Their time and memory costs can still be very different.

A: open clusters and scan their documents. For the random-bit example, use the first h bits as the cluster key. There are $C = 2^h$ possible clusters. Probing all keys through radius t gives $P = H(h, t)$ clusters and exactly the candidate rule analyzed above. With $h = 4$ and $t = 1$, scanning five clusters returns an expected 80 documents and has expected top-one recall 89.60%. The ratio $P/C = 5/16$ describes the expected fraction of documents read; it is not the nearest-document recall.

A k-means router uses distances to learned cluster centers instead of counting prefix errors. For each true neighbor, let J be the rank of its assigned cluster in the query's routing order. Probing the first P clusters retains that neighbor when $J \leq P$. Routing recall is therefore

$$r_{\text{route}}(P) = \Pr(J \leq P),$$

where the probability averages over queries and a uniformly selected member of each query's true top K . This identity defines what to measure. Cluster count and average cluster size alone do not supply the distribution of J .

B: follow bitplane branches. Consider a four-level tree where a path specifies the first four bits. Keep every path with at most one disagreement with the query. It reaches the same five keys as the example above, so its recall is also 89.60%. If every allowed path is represented, the

tree has $1 + 2 + 3 + 4 + 5 = 15$ nodes, including the root and five leaves. Some nodes may have no documents and can be skipped.

This example counts a particular complete set of paths. The actual best-first algorithm has a different stopping rule. It chooses a pending node using its score bound, splits on a query coordinate, and may stop when the node budget is used. Its candidate set depends on the weights, occupied leaves, leaf threshold, current result threshold, and visit order. A budget of 15 nodes does not mean that it reaches the five prefix keys above. To estimate its recall, run that selection process on sampled data or derive its visited paths for a small explicit tree.

A random exploration probability changes which pending node is chosen. At a fixed budget it can recover a missed neighbor or spend work on an unhelpful path. Its effect must be checked with the same queries and budget. Complete traversal with valid bounds is the exact control.

C: try keys near the query’s desired pattern. With equal weights, opening every short key through a Hamming radius gives Equations (43) and (47). With unequal weights, keys must be ordered by the sum of the weights of their wrong bits. Several low-weight mismatches can precede a single high-weight mismatch. The weighted distribution in the next subsections replaces the simple mismatch count.

For full keys, an exact weighted order and a valid stopping rule give exact local binary results. Short keys omit part of the score, so stopping after enough postings have arrived can lose neighbors. It can also achieve high recall when the prefix strongly predicts the full ranking. The calculation must account for both possibilities.

Literal prefix relaxation is another concrete rule. For query key 0000, replacing the last constraint by a wildcard accepts both 0000 and 0001. A prefix trie can retrieve that parent range, or only the new sibling range if the first key was already read. Flipping the last bit instead visits only 0001. Relaxing arbitrary query-selected coordinates is not a single ordinary prefix-trie parent step. The candidate sets and data access costs must follow the chosen structure.

Example: Conditional acceptance across two checks

Keep $d = 16$ and suppose we know the nearest document has two errors. A router that demands the first two bits match retains it with probability $\binom{14}{2}/\binom{16}{2} = 91/120$. Given that it passed, demanding the next two bits match has probability $\binom{12}{2}/\binom{14}{2} = 66/91$. Their product is $66/120 = 55\%$.

Multiplying $91/120$ by itself would give a different answer. Once the first two bits are known to match, the two errors must lie among the remaining fourteen positions. Knowing the first check passed changes the second calculation. The same issue arises when routing and lookup use overlapping bits.

6.10 Omitted-score bounds

Split the full score into retained and omitted coordinates:

$$S = T_h + Z_h, \quad T_h = \sum_{i \in I_h} q_i b_i, \quad Z_h = \sum_{i \notin I_h} q_i b_i. \quad (49)$$

Because $b_i \in \{-1, +1\}$,

$$|Z_h| \leq A_{\text{tail}} = \sum_{i \notin I_h} |q_i|. \quad (50)$$

Two documents can reverse their prefix order if the omitted contributions overcome their prefix score difference. A prefix gap greater than $2A_{\text{tail}}$ is sufficient to preserve that pair's order. This is a sufficient bound, not a typical-error estimate.

A query-specific check compares the best unseen prefix score plus A_{tail} with the worst fully scored retained result, as in Equation (31). This condition depends on the actual query. If the bound is too loose, more keys must be visited. Returning a fixed candidate count without checking that condition remains approximate.

When we omit coordinates from float vectors, the score error obeys the Cauchy–Schwarz inequality:

$$|q_{\text{tail}}^\top x_{\text{tail}}| \leq \|q_{\text{tail}}\|_2 \|x_{\text{tail}}\|_2.$$

Omitting coordinates and converting values to binary are separate approximations. Matryoshka training can improve the usefulness of prefixes, but it does not set either bound to zero or supply a universal recall percentage.

6.11 Weighted mismatches

Let E_i be one when document bit i disagrees with the query and zero otherwise. With $a_i = |q_i|$, the penalty is $L = \sum_i a_i E_i$. The score is $\sum_i a_i - 2L$, so ranking by largest score is the same as ranking by smallest L .

For query magnitudes $[4, 3, 2, 1]$, one error in the first coordinate costs 4. Errors in the last two cost $2 + 1 = 3$. The two-error document ranks higher. Counting wrong bits alone would reverse that order.

Under independent mismatch probabilities p_i , a particular error pattern e has probability

$$\Pr(E = e) = \prod_{i=1}^d p_i^{e_i} (1 - p_i)^{1-e_i}. \quad (51)$$

For a small dimension, list the 2^d patterns, calculate each full penalty and prefix penalty, and add their probabilities into the corresponding score groups. That directly constructs the joint distribution needed for recall: the probabilities of each full and prefix penalty pair. It also works with dependent bits if we supply their actual joint pattern probabilities instead of the product above.

To handle a finite top K , group patterns by each distinct full penalty ℓ . Let $p_{\leq \ell} = \Pr(L \leq \ell)$ and $p_{< \ell} = \Pr(L < \ell)$. For a fixed candidate acceptance event A , the same reasoning as the equal-weight case gives

$$\mathbb{E}[\text{Recall@ } K] = \frac{1}{K} \sum_{\ell} [G_K(p_{\leq \ell}) - G_K(p_{< \ell})] \Pr(A \mid L = \ell). \quad (52)$$

Each bracket counts expected top- K rows at that penalty. The conditional probability says what fraction pass, given that full penalty. This is exact for independently sampled documents from the stated pattern distribution, with an ID tie rule independent of the sampled patterns. It does not require independence *between bits*. It does require the joint distribution to be known.

Example: Balanced bits can still be dependent

Make eight-bit documents by sampling four balanced bits and repeating them in the last four positions. Every individual bit is still zero or one half the time. However, only sixteen of the possible 256 codes can occur. The independent-bit model assigns probability to 240 impossible codes. A correction to the arithmetic cannot repair that assumption; the pattern probabilities must change. Section 7.4 checks this example explicitly.

At hundreds of dimensions, listing every pattern is impractical. We can sample from a stated model, use a suitable dynamic program, or make an approximation to the score distribution. The required calculation determines the appropriate method. The small exact model remains useful for checking whether a faster calculation is answering the same question.

6.12 Independent-bit score correlation

The following models initially exclude routing. When routing is added, retain the same global true-neighbor event or apply Equation (48); replacing it with a different local top- K event changes the question. When computing correlation, assume the scores have nonzero variance, meaning they do not all have the same value.

Hold one query q fixed. Assume document signs are independent and equally likely to be -1 or $+1$. Then

$$\mathbb{E}[S \mid q] = 0, \quad \text{Var}(S \mid q) = \sum_{i=1}^d q_i^2, \quad (53)$$

$$\mathbb{E}[T_h \mid q] = 0, \quad \text{Var}(T_h \mid q) = \sum_{i \in I_h} q_i^2, \quad (54)$$

$$\text{Cov}(T_h, S \mid q) = \text{Var}(T_h \mid q). \quad (55)$$

Correlation measures how strongly two scores vary together. Under these assumptions, it is exactly

$$\rho_q = \text{Corr}(T_h, S \mid q) = \sqrt{\frac{\sum_{i \in I_h} q_i^2}{\sum_{i=1}^d q_i^2}}. \quad (56)$$

Keeping much of the score variance does not mean the omitted score has a small worst-case bound. Many small omitted coordinates can have a modest sum of squares but a large sum of absolute values. A prefix can therefore give useful approximate results while still needing many checks to prove they are exact.

Retaining a fraction v of score variance gives correlation $\sqrt{v}$, not recall v and not recall $\sqrt{v}$.

Example: Prefix recall under two random-data distributions

With equal coordinate variances, a 24-coordinate prefix of a 256-coordinate query retains roughly 24/256 of expected score variance. If instead the first 24 coordinates are constructed to hold 80% of expected score variance, their score varies much more closely with the full score. The controlled measurements in Section 7 test both cases. The second is a deliberately favorable distribution, not the definition of Matryoshka embeddings.

If query coordinates are random normals $q_i \sim \mathcal{N}(0, \sigma_i^2)$, we can choose each coordinate's variance σ_i^2 . The ratio of *expected* prefix and full variances is $\sum_{i \in I_h} \sigma_i^2 / \sum_i \sigma_i^2$. For a specific sampled query, use its actual Equation (56). The expectation of a ratio need not equal the ratio of expectations.

6.13 Dependent document bits

A covariance matrix records how coordinates vary together. Its diagonal entries are individual variances. An off-diagonal entry describes whether two coordinates tend to move together or in opposite directions. Independent, balanced signs have unit diagonal entries and zero off-diagonal entries.

For a general binary document vector with mean μ and covariance matrix Σ , let q_h be q with omitted coordinates set to zero. Then

$$\rho_q = \frac{q_h^\top \Sigma q}{\sqrt{(q_h^\top \Sigma q_h)(q^\top \Sigma q)}}. \quad (57)$$

This includes dependencies between coordinates. Restricting documents to a cluster or filter can change both μ and Σ . How coordinates vary together across the whole database can differ from how they vary inside the query's selected clusters.

A learned prefix may preserve ranking information through these dependencies even when its share of the sum of squared coordinate values is small. Conversely, a prefix can have a large sum of squared values and still fail to identify the extreme nearest neighbors. Estimating covariance also does not prove that the rare highest scores follow a normal distribution.

6.14 Nonuniform mismatch probabilities

Let $X_i = \mathbf{1}[b_i \neq t_i]$. If the X_i are independent with the same disagreement probability p , and all relevant weights equal a , then a mismatch count over h coordinates satisfies

$$M_h = \sum_{i=1}^h X_i \sim \text{Binomial}(h, p), \quad \pi_h = aM_h. \quad (58)$$

A Hamming radius r visits

$$V_h(r) = \sum_{u=0}^r \binom{h}{u} \quad (59)$$

keys, and its expected candidate count is

$$N_* \Pr(M_h \leq r) = N_* \sum_{u=0}^r \binom{h}{u} p^u (1-p)^{h-u}. \quad (60)$$

Here N_* is the relevant reference pool size; it is N without filters or cluster restriction. Equation (60) is a candidate-count model, not yet recall.

If $p = 1/2$, every key has equal probability and the expected count reduces to $N_* V_h(r)/2^h$. Different p_i with equal weights give a Poisson-binomial count: a sum of independent zero-or-one outcomes with different probabilities. Unequal weights give a weighted Bernoulli sum, which also multiplies each outcome by its weight. Neither is described by one ordinary binomial with an average p .

6.15 Recall under penalty thresholds

Write $F_X(x) = \Pr(X \leq x)$ for a cumulative distribution function (CDF): the probability that X is at or below a threshold.

Let A_h be the prefix penalty and B_h the omitted-coordinate penalty. Full penalty is $A_h + B_h$. Under coordinate independence, the two sums are independent. Let u be the full-penalty threshold for a true neighbor, and let v be the candidate prefix-penalty threshold. Then

$$r_{\text{population}}(u, v) = \frac{\Pr(A_h \leq v, A_h + B_h \leq u)}{\Pr(A_h + B_h \leq u)} = \frac{\int_{a \leq v} F_{B_h}(u - a) dF_{A_h}(a)}{F_{A_h + B_h}(u)}. \quad (61)$$

The numerator is a weighted sum. For each allowed prefix penalty, it multiplies the probability that the remaining coordinates satisfy the full-score threshold by the probability of that prefix penalty, then sums the contributions.

A candidate-count calculation alone omits the second condition: a retrieved document counts toward recall only if its full penalty is also low enough to qualify.

For equal weights and an integer full radius r_d , the formula becomes

$$\frac{\sum_{u=0}^{\min(h, r_h, r_d)} \binom{h}{u} p^u (1-p)^{h-u} F_{\text{Bin}(d-h, p)}(r_d - u)}{F_{\text{Bin}(d, p)}(r_d)}. \quad (62)$$

This is exact for the stated population events. Choosing those radii to approximate top K and a candidate budget is another step. We must also account for ranks in a finite sample and score ties.

Example: Candidate fraction is not recall

Use all 64 six-bit codes as documents and an all-positive equal-weight query. The best seven codes have zero or one mistake. Keep only codes matching the first two query bits: 16 codes pass, so the candidate fraction is $16/64 = 25\%$. Five of the true seven pass: the ideal code and the four codes whose one error is in an omitted coordinate. Recall is $5/7$, about 71.4%, not 25%. Allowing one error in the two-bit prefix retrieves 48 codes and includes all seven. Both numbers follow by counting the same true-neighbor event.

6.16 Weighted key counts

For a fixed query and lookup coordinates I_h , define

$$V_q(v) = \# \left\{ S \subseteq I_h : \sum_{i \in S} |q_i| \leq v \right\}. \quad (63)$$

This counts every attempted key through threshold v , whether occupied or empty. Under independent uniform keys, expected candidates are $N_* V_q(v)/2^h$. With real key probabilities,

$$\mathbb{E}[F(v) \mid q] = N_* \sum_{k: \pi_q(k) \leq v} \Pr(\text{key} = k \mid q, \text{cluster}, \text{filter}). \quad (64)$$

For a clustered lookup, apply Equation (64) separately with pool size e_j and the key probabilities within cluster j , then sum the results.

An average bit-disagreement rate does not determine the probabilities of complete bit patterns. The empty-key count V_q and occupied candidate count F are separate inputs to the latency formula. A dense document distribution near the query key can make F large while V_q remains small; a sparse key space can do the reverse.

For small h , split weights in two, enumerate each half's subset sums, sort one list, and count pairs whose sum is at most v . This meet-in-the-middle calculation uses roughly $O(2^{h/2})$ memory and $O(h2^{h/2})$ sorting/search work. It is an analysis tool for counting keys, not the measured latency of an online hash enumerator.

For weights a_i rounded to nonnegative integers, a dynamic program builds the probability of each penalty from the previous step:

$$f_i(s) = (1 - p_i)f_{i-1}(s) + p_i f_{i-1}(s - a_i). \quad (65)$$

Replacing probabilities by counts and adding the two counts from the previous step computes the number of keys. This is exact for the quantized-weight model. It is not an exact calculation for unquantized query weights. If score quantization incurs a uniform score error at most ϵ , a score gap greater than 2ϵ is sufficient to preserve a pair's order. Otherwise candidates near the boundary need exact scoring or a safe expanded bound.

6.17 Normal score approximation

When a score adds many small independent contributions, a normal distribution can sometimes approximate its shape. Recall requires a joint distribution: how the full score and prefix score vary together. This is an additional assumption, not a consequence of knowing their correlation. Subtract each score's mean and divide by its standard deviation. This puts them on the same scale as (Z, Y) with correlation ρ . For a true fraction $a = K/N_*$ and selected candidate fraction f , set

$$z_K = \Phi^{-1}(1 - a), \quad y_f = \Phi^{-1}(1 - f),$$

where Φ and ϕ are the standard normal CDF and density. Under the joint-normal assumption, Y , when we know $Z = z$, has mean ρz and variance $1 - \rho^2$. Its probability of exceeding the prefix threshold is therefore $\Phi((\rho z - y_f)/\sqrt{1 - \rho^2})$. Multiply by the full-score density $\phi(z)$, add across all scores that qualify as true neighbors, then divide by their probability a . The implied recall is

$$\hat{r} = \frac{1}{a} \int_{z_K}^{\infty} \phi(z) \Phi\left(\frac{\rho z - y_f}{\sqrt{1 - \rho^2}}\right) dz. \quad (66)$$

We average over *all* scores that qualify as true neighbors. Substituting only $z = z_K$ calculates the passing probability at that one boundary, not the average over all qualifying scores. When $\rho = 0$, recall is f . In the limit of identical rankings, $\rho = 1$, it is $\min(1, f/a)$, subject to ties.

This formula is not valid simply because an embedding is called Matryoshka. A short weighted binary prefix has only a finite set of possible scores. The probabilities of its highest scores can differ greatly from those predicted by a normal distribution. At a billion documents, top- K events may be much rarer than anything covered by a small validation sample. Scores that vary closely together overall do not prove 99% recall.

The controlled study below estimates score correlation on one query group and checks predicted recall on different queries. Even there, the real-data normal approximation misses by several percentage points. Such a model can suggest explanations for further testing; it cannot establish which method is fastest at 99% recall on real data.

6.18 Implications for the three methods

- **A:** exact local scanning has no local candidate-search loss; global loss comes from routing, candidate/reranker score changes, and the declared filtering task.
- **B:** exact bound-based traversal has the same binary result as A on the same selected pool. Finite budgets require query-level measurements or a distribution model for visited nodes and surviving true neighbors. A node count alone is not enough.
- **C:** full keys with complete weighted ordering are exact locally. Short keys can be accurate when their ranking preserves true neighbors and the candidate budget is sufficient. This possibility requires measurement. Their enumeration and posting costs remain separate from the recall benefit.

A useful estimate needs a stated candidate rule and a distribution that fits the relevant neighbors. The checks below show both a case where the exact calculation works and a case where a plausible assumption produces an incorrect prediction.

7 Empirical evidence

These measurements assess the analytical reasoning. The experiments cover different stages of search; comparisons therefore require explicit attention to their measurement scope.

7.1 One-million-passage study

The archived study used one flat cluster, a fixed 256-bit score, 100 binary candidates, leaf size 32, deterministic branching, cached query embeddings, and one CPU thread. There were 100 validation queries for selecting configurations and 200 separate test queries, with three timing repeats. It covered pools from 1,000 to one million real MS MARCO passages.

One-million-document test	Median latency	Binary recall	Fully scored rows
Packed scan	25.14 ms	100%	1,000,000
Branch, budget 32,768	244.98 ms	99.26%	about 163,766
Unlimited branch	1,068.84 ms	100%	about 999,961

Branching reduced scoring under the finite budget but was slower at matched target recall. Across the study, branching was not faster at either the median or p95 among the tested 95% and 99% target settings. This conclusion is limited to the implementation, dataset, hardware, and tested settings. It does not establish that all bitplane representations are slower.

The original implementation uses dense active masks for every node. Its measured split-word counter omits leaf reads, which must be derived from the visited-node counts and cluster widths. The physical mask size at one million rows is $15,625 \times 8 = 125,000$ bytes. A queue with 20,000 such masks would contain 2.5 GB of masks alone. This example explains how query RAM can be much larger than the bytes used to store the packed codes; it is not a claimed observed queue size.

The source summary is `results/scaling-msmarco-1m/analysis/settings.csv`. Full run details and the checking procedure are stored beside it. The report’s verification script checks the numerical rows quoted here.

7.2 FiQA search study

The expanded FiQA study used 57,638 documents and 648 queries, with 583 settings including random seeds. Its timings include routing, binary candidate search and float reranking, with embeddings cached. Its quality measures include float Recall@10 and nDCG@10, rather than the scaling study’s binary Recall@100.

Method	Probes	Node budget	Median ms	Float Recall@10
Sign-route binary scan	16	–	0.7524	0.6784
Sign-route branch	32	2,048	0.6256	0.7006
Sign-route branch	64	2,048	0.6466	0.7062
Float IVF reference	8	–	0.1661	0.9127

These selected examples show that probing more clusters with limited work inside them can be a useful tradeoff between routing coverage and local search work. They do not measure the effect of branching alone: the selected pools differ. Within the same selected buckets, unlimited branching was slower than scanning. The float IVF row uses a different scorer and was a strong practical reference in that experiment.

Random exploration at probability 0.1 did not show a consistent improvement. Its purpose is to change visit order; it does not provide a mathematical guarantee of higher recall. The full FiQA

report retains unsuccessful configurations, empty returns, and the relevant measurement scope. Its source is `results/fiqa-2026-09-09-dense/README.md` and the accompanying data.

7.3 24-bit prefix evaluation

The small quality study used the same full weighted-binary score and row-ID tie rule for all methods. There was no outer routing: $C = P = 1$. Two synthetic datasets had 20,000 random binary documents. The real-data case used the first 100,000 rows of the cached Nomic/MS MARCO sample. All cases used $d = 256$ and $h = 24$.

One synthetic query distribution gave all coordinates equal variance. The other multiplied the first 24 coordinates so that they held 80% of *expected* full-score variance. It was a controlled favorable case for a prefix, rather than a general model of Matryoshka behavior.

For each case, 16 queries estimated a score correlation and 16 different queries checked the normal-model prediction. The native full scan, which runs the compiled search code, agreed with an independent exact score sort on all 96 query references. The short-key candidate set included all documents in keys tied at the final prefix threshold, then used the exact full score to keep the top 100.

Data, candidate target	Measured recall	Normal prediction	Gap	Median keys
Random equal variance, 1,000	23.00%	21.84%	−1.16 pp	846,156
Strong prefix, 1,000	95.00%	97.01%	+2.01 pp	846,156
Cached Nomic, 1,000	25.25%	17.41%	−7.84 pp	6,976
Cached Nomic, 5,000	52.13%	42.57%	−9.56 pp	35,504
Cached Nomic, 10,000	66.19%	58.46%	−7.73 pp	78,370

“Gap” is prediction minus measurement in percentage points. Values are rounded from the per-query evidence. The identical synthetic key counts are expected: the strong-prefix case scales all first-24 query weights by the same factor, which changes neither their key order nor the prefix candidate sets. It changes their importance in the *full* score.

The constructed favorable case reaches 95% recall with 1,000 candidates, demonstrating that a short prefix can retain high recall under a suitable distribution. However, the statistical model differs from measured real-data recall by several percentage points. Errors of this size prevent it from establishing that a 99% recall target is met.

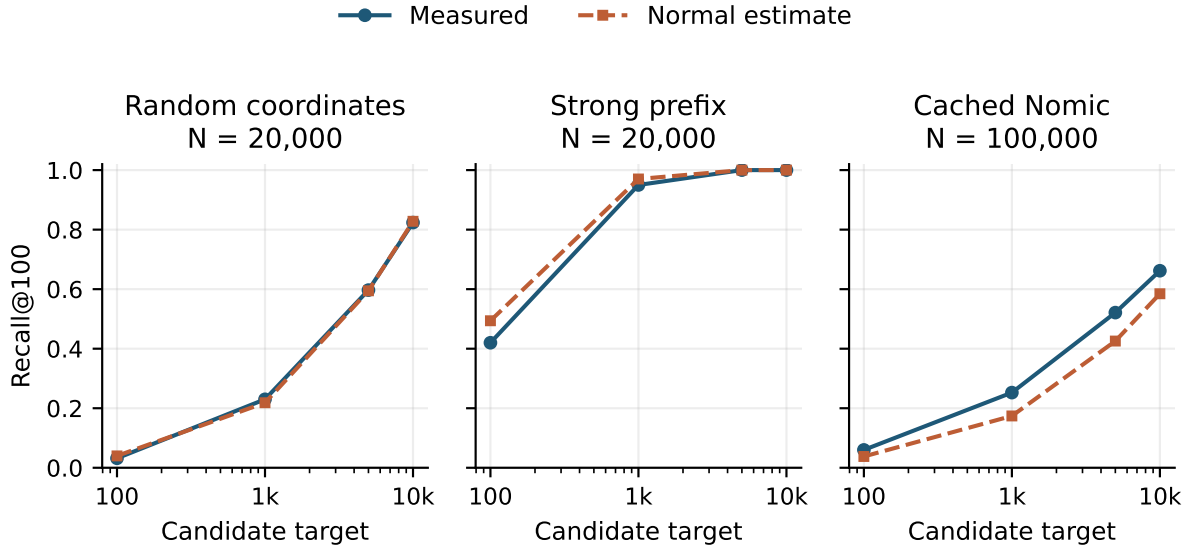


Figure 7: Measured short-key recall versus the checked normal approximation. Each panel states which document set it uses.

The key counts were calculated from the actual query weights by splitting the weights in two, listing each half’s subset sums, and counting qualifying pairs, including empty keys. The candidate set was constructed by calculating every document’s prefix score for quality evaluation. This computation evaluates candidate quality rather than the latency of an implemented hash search. Its wall time therefore cannot be reported as the latency of Approach C.

In the same favorable synthetic case, native B at budget 512 recovered about 99.38% of the binary top 100 while fully scoring about 4,122 documents on average. In the real-data case, B reached about 58.06%, 86.56% and 99.56% at budgets 512, 2,048 and 8,192. These are quality and work observations for a flat pool with a 1,000-candidate heap. They do not establish a latency ranking between an implemented native B and a hypothetical C.

7.4 Recall formula validation

The small arithmetic validation script lists every ordered document set and sorts its documents directly. This includes duplicate codes, score ties, empty prefixes, full prefixes, $K = 1$, and $K = N$. All twenty settings agree with Equation (47) to numerical precision. The 16-bit worked example is then calculated from the formula rather than by enumerating every possible corpus.

A second check uses eight-bit documents, $N = 64$, $K = 5$, and the first three bits as lookup coordinates. A document passes when at most one of those three bits disagrees with the query. For each distribution, the script calculates the exact expected recall by grouping all possible patterns by full penalty, then samples 10,000 independent corpora and checks the returned top five directly. The query has positive coordinates with the following magnitudes:

- Equal weights: $[1, 1, 1, 1, 1, 1, 1, 1]$; all bits independent and balanced.
- Strong prefix: $[4, 3, 2, 1, 1, 1, 1, 1]$; the same independent document distribution.
- Weak prefix: $[1, 1, 1, 4, 3, 2, 2, 2]$; the same independent document distribution.
- Paired bits: all query magnitudes one, with the last four document bits repeating the first

four. Only sixteen patterns can occur.

These deliberately simple assumptions allow one factor to change at a time. The resulting distributions do not model a particular embedding system.

Case	Exact stated model	Sampled mean	Assume independent bits
Equal weights	93.847%	93.784%	93.847%
Strong prefix	99.999%	100.000%	99.999%
Weak prefix	74.248%	74.430%	74.248%
Paired bits	$\approx 100\%$	100.000%	93.847%

The candidate rule is identical in every row. It returns an expected 32 of the 64 documents, but its recall changes with the score weights and document distribution. The last row shows the effect of an incorrect assumption: individual bits remain balanced, yet assuming they are independent underestimates recall by about 6.15 percentage points.

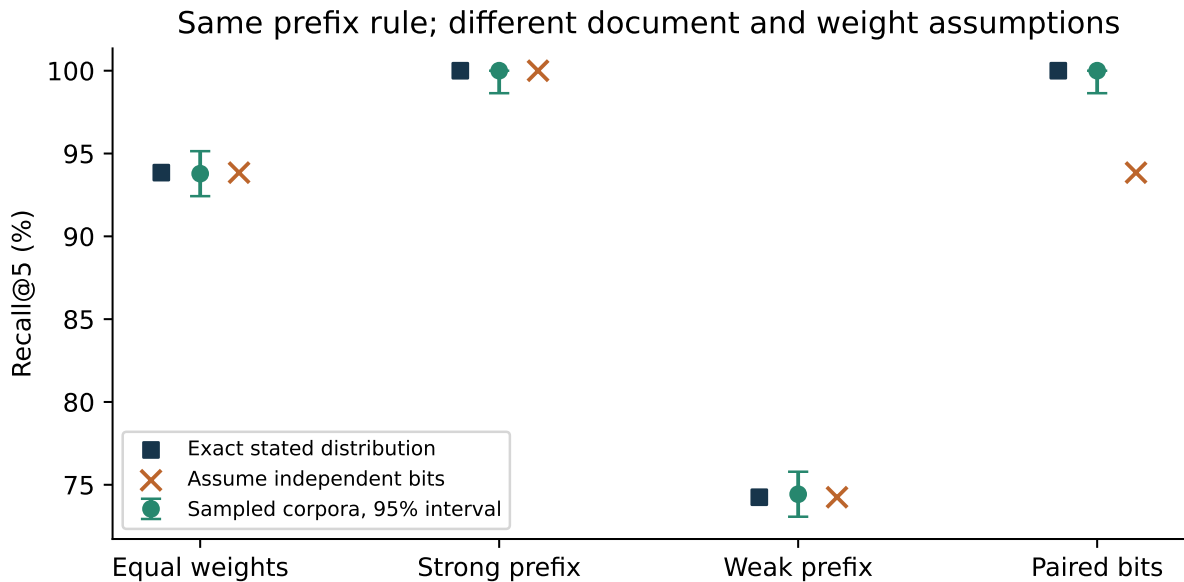


Figure 8: Exact calculations and independently sampled corpora agree for the stated distributions. The paired-bit row shows the error from using the wrong distribution.

Each sampled corpus contributes one recall value in $[0, 1]$. A conservative uncertainty bound for their mean follows from

$$\Pr(|\bar{r} - \mathbb{E}[r]| \geq \epsilon) \leq 2e^{-2n\epsilon^2}.$$

Setting the right side to 0.05 gives $\epsilon = \sqrt{\log(40)/(2n)}$, about 1.36 percentage points at $n = 10,000$. The plotted intervals use this bound and are clipped to $[0, 1]$. This remains meaningful when no misses were observed. A zero sample variance by itself would produce a misleading zero-width normal interval near 100%. The bound assumes independent sampled corpora; the ten thousand simulations do not establish recall for real embeddings.

For the first twelve sampled corpora in each case, the check also calls the current native scan and bitplane search. All 48 full scans and all 48 unlimited branch searches match the independently sorted top five. With a 16-node budget and leaf size four, mean branch recalls are 80.00%, 98.33%, 100.00%, and 85.00% in the table’s order. At exploration probability 0.1 and the same

budget, they are 75.00%, 98.33%, 100.00%, and 85.00%. These small examples show that the budget and weights affect which neighbors survive. They are not estimates of performance on a large dataset or evidence that random exploration always reduces recall.

The script separately reaches the same fixed prefix-radius candidate set through key grouping and through the union of bitplane leaves. All 48 candidate sets agree exactly. This checks the shared-set argument; it does not claim that budgeted best-first traversal visits those same leaves. No latency ranking is inferred from these quality checks.

7.5 Prediction uncertainty

The earlier study showed discrepancies between the normal approximation and measured recall. To assess variation across queries, the reanalysis draws 10,000 sets of sixteen queries from the sixteen evaluation queries, allowing a query to appear more than once in each set. For each resample, it averages the paired difference $\hat{r}(q) - r(q)$. The middle 95% of these averages defines the bootstrap interval below. A percentage point is an absolute difference: predicting 30% when the observation is 25% is an error of +5 points.

Case	Candidate target	Mean error (points)	95% interval
Random signs	1,000	-1.16	[-3.72, 1.41]
Strong prefix	1,000	+2.01	[0.08, 4.14]
Cached Nomic	1,000	-7.84	[-14.47, -1.54]
Cached Nomic	5,000	-9.56	[-18.25, -0.81]
Cached Nomic	10,000	-7.73	[-16.17, 0.64]

This reanalysis uses previously examined queries rather than a new test set. The estimated score correlation is held fixed, so the intervals do not include uncertainty from estimating that correlation. They also do not cover corpus changes, model changes, or selection among many configurations. When every checked query gives the same outcome, bootstrap variation can vanish; that is not proof that future queries have no variation.

A proposed requirement of at most two points of mean error fails on every cached-Nomic setting shown here. Even the random-sign row, whose observed mean error is small, has an interval that extends beyond that tolerance. The normal model can therefore explain a possible mechanism, but it cannot certify 99% recall on real data.

Improving the prediction requires new evidence about the relevant score distribution. For example, estimate how full and prefix penalties vary together on queries used to fit the model, fix the candidate rule, and check it on separate queries. If the distribution comes from samples, report how much detail those samples provide about the rare nearest scores. A small sample can miss rare but important patterns; moving from 100,000 to a billion documents makes that issue more severe. The current extension does not claim a newly validated real-data predictor.

7.6 Limitations

The study does not establish:

- real recall at a billion documents;
- a routing-recall model across arbitrary cluster counts;
- a normal distribution of the highest scores for every query or a fixed percentage of Matryoshka prefix information;
- a concrete hash enumerator’s timing, peak queue size, or SIMD speed;
- float-vector recall, human relevance, or final answer accuracy from binary recall alone.

Float-vector top-100 results and binary-score top-100 results answer different questions. This report keeps their comparisons separate. A method that searches binary scores exactly can still miss neighbors under the original float-vector score.

7.7 Further evaluation

Further evaluation should fix the representation, query set, filtering task, and reference score before tuning. Configuration selection and evaluation should use separate query sets. For every method, the record should retain failures and empty returns, together with work counts and timings. For C, implement the actual enumerator before claiming its measured latency. For B, record split words, leaf words, documents scored and peak memory for pending nodes separately.

At each target recall, compare measured latency percentiles using configurations that actually reach that target. Use the same query population and report the uncertainty of quality estimates. Query-level resampling is appropriate; counting repeated timings as independent quality samples is not. The measured cases provide evidence about specific mechanisms; they do not characterize every data distribution.

8 C++ comparison

The earlier studies had two limitations: the branch counter omitted leaf-mask reads, and the short-key recall experiment did not measure a C++ key index. This study implements all three methods with the same score and checks their components before timing complete requests. These results are distinct from the historical results reported earlier. In this study, native refers to the C++ implementation called by the Python driver.

8.1 Shared setup

A, B and C use the same C++ lookup-table scorer and the same top- K heap. Each query remains floating point; each document stores one sign bit per coordinate. The final score is the full weighted binary score from Section 1, with smaller document IDs first when scores tie. There is no float reranker in this comparison. Thus recall means recovery of the exact binary-score top 100, not relevance to a person reading the documents.

The Python driver supplies cached query vectors and selected cluster IDs. A fresh process builds one configuration, releases the mapped input files, runs a warmup query, then measures routing

and search together. The C++ search also records its own time. The outer timer includes the work of calling C++ from Python; the C++ timer alone would leave that out.

The machine is an Apple M5 Max with 128 GiB of physical memory. Each worker uses one CPU thread and handles one query at a time. The comparison budget is 32 GB, in decimal bytes. The machine’s physical capacity and the experiment’s memory constraint are different numbers. Power configuration is recorded before and after each worker. Equal settings do not establish fixed CPU frequency or a completely idle machine, so timings are repeated in separate processes.

8.2 Key storage

The C++ key-index build sorts rows by key inside each cluster and stores their full codes in that same order. Its existing document IDs are the postings. There is no extra N -element array of posting references. This is the layout alternative described after Equation (30).

The directory is a C++ `unordered_map`. It uses separate chaining: each stored entry has its own node in memory, and an array of pointers leads to those nodes. Equation (29) describes a different, open-addressed layout and must not be used as its measured allocation. Let L_j be the number of hash buckets, b_{node} the allocated node size including fields and alignment, and b_{ptr} a bucket-pointer size. Then the implemented allocation is

$$M_{C,\text{allocated}} = M_{\text{codes,capacity}} + M_{\text{IDs,capacity}} + \sum_j (U_j b_{\text{node}} + L_j b_{\text{ptr}}) + M_{\text{containers}} + M_{\text{allocator}}. \quad (67)$$

The last two terms cover the objects that hold the data and the records used to manage memory. Their sizes depend on the standard library and build. Payload counts only the stored fields, excluding that overhead and unused capacity. For the directory, each occupied entry holds a key, the start of its row range, and the number of rows:

$$M_{\text{directory,payload}} = \sum_j U_j (b_k + 2b_o).$$

In this build those fields occupy 4, 8 and 8 bytes respectively. Their 20-byte sum is a payload count, not a claim that an allocated hash node occupies only 20 bytes.

B retains packed document rows and a second, transposed bitplane representation. A is built without bitplanes, so its memory count excludes that allocation. All three retain the IDs required to return original document rows.

8.3 Key search

The implementation has the same free key coordinates in every selected cluster. It enumerates one weighted sequence of keys and looks up each key in every selected cluster. This differs from the per-cluster target example earlier in the report.

Listing 13: The native key-search policy used in the measured study.

```

SearchKeys(query, selected_clusters, K, candidate_target, lookup_limit):
    build the shared full-score lookup table
    order the key coordinates by increasing absolute query weight
    queue = [the ideal key, with mismatch penalty zero]
    best = empty top-K heap
    while queue is not empty:
        if best is full and the best remaining score bound cannot improve it:
            stop
        state = remove the lowest-penalty flip set
        key = ideal key XOR state.flip_mask
        for cluster in selected_clusters:
            if the lookup limit is reached: return best
            count one lookup, including an empty lookup
            for document in directory[cluster][key]:
                score all document coordinates; offer the score and ID to best
            if a nonzero candidate target is reached: return best
        add the next flip sets to the queue
    return best in score/ID order

```

The score bound is the full ideal score minus twice the key’s mismatch penalty. Coordinates outside the key are allowed their best possible contribution. This remains a safe upper bound when routing fixes some of those coordinates, although it may be loose. The implementation does not yet tighten it separately for each cluster.

The candidate target is checked after finishing a key in all selected clusters. A lookup limit can stop in the middle of that cluster loop. Both limits are approximations; zero disables a limit. The exact score bound remains available in either case. Equal-score candidates require the shared ID rule, so the bound includes a small allowance for floating-point rounding.

Because every cluster uses the same key order, one queue serves the search, rather than P separate queues. The work counts include queue preparation and generated states once, and each cluster lookup separately. Every returned candidate still receives all G score-table terms. The tail-only and full-key score reuse discussed earlier are possible alternatives, but they are not credited to the measured implementation.

8.4 Work counts

Independent Python formulas check the bytes required for code, ID, plane and key-directory data. The report also records C++ array capacity: an array may reserve space for more entries than it currently holds. Separate tests compare returned IDs against an independent full-score sort. The reference calculation starts from document signs, not from the C++ packed-code scorer.

B reports split-word visits and leaf-word visits separately. C reports generated keys, directory lookups including misses, and fully scored documents. Both report their largest tracked queue or mask allocation. These counts explain a timing result; a low document-score count alone does not establish that the whole query performed little work.

Example: Two bitmap cuts

The corpus contains every possible eight-bit document once, so $N = 256$ and $W = \lceil 256/64 \rceil = 4$ words. The first three query coordinates have magnitude one; the remaining five have magnitude 0.0001. The search requests four results and stops splitting when 64 documents remain.

Two matching cuts leave $256/2^2 = 64$ documents. The search visits the root, its preferred child, and the leaf: three nodes. It reads $2 \times 4 = 8$ split words and another four words to enumerate the leaf. It then fully scores 64 documents. At the second split, one queued alternative, the parent and both new children occupy four masks, or $4 \times 4 \times 8 = 128$ bytes. The native measurement matches each count and returns the independently calculated top four. With a one-node budget it returns no candidates; this truncated control confirms that the test detects missing results. This example establishes the stated path, not a universal bound of two cuts for arbitrary query weights.

8.5 Timing model

Operation counts do not have a fixed conversion to time. A lookup can reuse data in the CPU cache or fetch data from a new memory page, with different costs. The first timing model therefore fits a nonnegative cost to each of a few measured work counts. Its search-time inputs are:

- A: a fixed term and the number of full score-table terms.
- B: the same terms, split plus leaf word visits, and visited nodes.
- C: the same score terms, directory lookups, and a queue-work estimate.

Each fitted cost includes arithmetic, memory reads and result selection together. It does not measure the time of one CPU instruction. Measured routing time is added per query before taking a median; summing the medians of separate stages would not give the median complete request.

There were 117 calibration configurations on random sign data with one cluster. An initial fit minimized squared errors in milliseconds. The slowest queries had the most influence on that fit, leaving large percentage errors for some fast branch queries. Fitting squared percentage errors on the same data improved that check. The original fit remains saved.

The revised coefficients were then frozen and checked on 52 configurations at different sizes and a new seed. All four scan settings and all 24 key settings had median prediction error within 20%. For B, 21 of 24 settings met that criterion; its worst error was 27.2%. The tables retain the failures. These fitted costs apply to the checked sizes of data used during a query; they do not establish timing accuracy at a billion documents.

This model also uses *observed* work counts. Predicting how many branches or empty keys a new query will visit is a different problem, governed by the document and query distribution. A fit that converts measured work into milliseconds does not validate a forecast of recall or candidate count. The later tuning comparisons use measured latency, rather than allowing a poorly transferred model to choose a winner.

8.6 Baseline settings

The real-data study reuses 256-bit Nomic/MS MARCO document codes at $N = 10,000$, $100,000$ and $1,000,000$. All methods have access to the same candidate router families and cluster layouts; each may select its own fastest tested configuration at a target recall.

A coarse 100,000-document run initially suggested a B advantage near 95% recall. The scan baseline’s next available probe setting jumped from 92.76% recall to a full scan. Adding intermediate probe counts removed that apparent advantage, showing that it resulted from the tested probe counts rather than faster traversal by B.

The later probe choices use the actual ranks of the correct neighbors’ clusters. Let r_{qi} be the rank of the cluster containing correct neighbor i for tuning query q . For Q queries and top K , the exact local-scan recall after probing P clusters is

$$\hat{R}_{\text{route}}(P) = \frac{1}{QK} \sum_{q=1}^Q \sum_{i=1}^K \mathbf{1}[r_{qi} \leq P]. \quad (68)$$

Sorting these QK ranks gives the smallest empirical cutoff for any requested target. The actual router checks the proposed cutoff; tied cluster scores are handled explicitly. This addresses the observed gap in the tested probe counts; it does not imply that P/C determines recall.

8.7 New-query checks

The refined runs were combined into 6,821 distinct settings. Timing observations from equivalent configurations are combined. Random exploration uses a fixed set of seeds, averaged per query; the fastest lucky seed is not selected as the method’s result. The chosen parameters were frozen before evaluating 200 query IDs excluded from the original 300-query collection.

Two selection rules were retained. The *mean* rule picks the fastest measured setting whose tuning recall reaches the target. The *conservative* rule repeatedly draws samples from the existing tuning observations and requires a lower percentile of the resulting recall values to reach it. This is bootstrap resampling; it does not create new independent queries. This second rule leaves more margin, but it is still a rule fitted to a finite sample, not a guarantee about all future queries.

There are 72 distinct frozen settings. Their predeclared seeds, three process blocks and three timing repeats produced 258 workers and 154,800 query observations. All workers completed and kept the same recorded power configuration. The largest process lifetime peak was 231,129,088 bytes, below the 32 GB constraint for these measured runs. It includes construction and runtime memory, so it is more conservative than the logical index payload alone.

Only 7 of 36 mean-selected target points reached their requested recall on the new queries. All 36 conservative points did. Re-encoding three old query texts with the fresh-query recipe exactly reproduced their cached vectors; no parameters were changed in response to the misses. A tuning average of 99% does not guarantee 99% recall on another finite query sample.

<i>N</i>	Target	Median milliseconds			Achieved recall (%)		
		A	B	C	A	B	C
10,000	80%	0.0979	0.1052	0.1129	84.91	84.91	84.91
10,000	90%	0.1339	0.1474	0.1560	91.79	91.79	91.79
10,000	95%	0.1683	0.1818	0.1968	95.84	95.84	95.84
10,000	99%	0.2330	0.2542	0.2812	99.27	99.27	99.27
100,000	80%	0.2345	0.2454	0.2698	80.55	80.55	80.55
100,000	90%	0.5034	0.5293	0.5918	90.98	91.29	91.29
100,000	95%	0.7527	0.7976	0.9308	95.47	95.47	95.57
100,000	99%	1.6219	1.6779	1.8109	99.44	99.44	99.44
1,000,000	80%	0.6749	0.6985	0.7506	82.53	82.53	82.53
1,000,000	90%	1.6341	1.6856	1.8802	91.73	91.73	91.73
1,000,000	95%	3.3600	3.4295	3.7055	95.47	95.47	95.47
1,000,000	99%	9.5796	9.9645	11.0127	99.25	99.25	99.25

The table reports the conservative choices. Each method can choose different cluster counts, probe counts and local settings. “Target” is the common minimum requirement; their achieved recalls need not be numerically identical. Exact settings and all mean-rule target misses are saved alongside the measurements.

To estimate uncertainty, we repeatedly sample the recorded queries and whole process blocks, using the same draws for both methods in a pair. These draws reuse existing observations; they do not add new independent queries. Repeated timings also do not count as independent recall observations. A supported alternative speedup requires both settings to meet the target and memory/power checks, and the paired 95% interval for $t_A/t_{\text{alternative}}$ to lie above one. None of the 48 evaluated comparisons meets that rule. These intervals describe individual comparisons; they are not simultaneous guarantees across every setting.

The result applies to this comparison: for this cached binary score, data, single-thread implementation and tested parameter space, the independent evaluation did not establish a B or C speedup over the tuned scan baseline. It does not exclude different query weight distributions, other data layouts, hardware kernels or larger corpora.

8.8 Reproduction

Reproduction instructions are provided in `experiments/README.md`. The small configuration checks components without a dataset download. The isolated calibration and confirmation configurations check the timing model. The frozen real selection is stored in `results/frozen-real-2026-09-10/`; its independent measurements and uncertainty are in `results/independent-evaluation-2026-09-10/`. Each run saves its settings, hashes identifying the source and C++ build, environment, per-query observations and process status. Before archiving a ZIP of a case, every file is checked against the original.

The following files implement the corresponding parts of the study:

Paper topic	Code to inspect
Shared binary score	<code>cpp/score.hpp</code>
Full scan and bitplane traversal	<code>cpp/index.cpp</code>
Weighted key enumeration	<code>cpp/key_index.cpp</code>
One complete measured query	<code>experiments/worker.py</code>
Independent storage/work formulas	<code>experiments/models.py</code>
Pooled tuning choices	<code>experiments/choices.py</code>
Frozen-query replay and comparison	<code>experiments/evaluate.py</code> , <code>experiments/evaluation_report.py</code>

9 Known strong coordinates

The independent test on real embeddings did not establish an advantage for B or C. Controlled data with known high-weight coordinates allow the conditions for an advantage to be examined. This experiment evaluates algorithm behavior under an explicit distribution; it does not claim that trained Matryoshka embeddings follow that distribution.

9.1 Assumptions

Let each document coordinate independently have sign $+1$ or -1 with probability $1/2$. The signs are therefore independent and balanced. A query has m strong coordinates of magnitude a , and $d - m$ weak coordinates of magnitude $\epsilon > 0$. Its signs are independently random. Scaling the whole query to unit length does not change any ranking or the following inequality:

$$(d - m)\epsilon < a. \quad (69)$$

A single strong mismatch loses $2a$ score. Even changing every weak coordinate from its best to its worst sign loses only $2(d - m)\epsilon$. Therefore any document matching all strong coordinates outranks every document with a strong mismatch. More generally, fewer strong mismatches always outrank more strong mismatches; the weak coordinates decide the order inside one mismatch count.

The experiment sets $d = 256$, $m = 13$, $a = 1$ and $\epsilon = 0.0001$. The weak total is 0.0243, so Equation (69) holds. One setup places the strong coordinates first. The other selects a different set of 13 coordinates for each query. Neither changes the document distribution. All 40 generated queries are retained, with the first 20 used for tuning and the remaining 20 for checking frozen choices.

9.2 One-key recall

Let Z count documents matching all m strong signs. Under the stated independent distribution,

$$p = 2^{-m}, \quad Z \sim \text{Binomial}(N, p), \quad \mathbb{E}[Z] = N2^{-m}.$$

Suppose a method opens exactly that key, scores every document in it, and returns at most K . Every one of those documents outranks all documents outside the key. Its recall on this corpus is therefore exactly

$$R_{\text{ideal}} = \frac{\min(Z, K)}{K}. \quad (70)$$

To obtain expected recall over random corpora, observe that $\min(Z, K)$ counts one for every threshold $i \in \{1, \dots, K\}$ that Z reaches. Taking expectations gives

$$\mathbb{E}[R_{\text{ideal}}] = \frac{1}{K} \sum_{i=1}^K \Pr(Z \geq i). \quad (71)$$

This finite binomial calculation retains variation in Z , rather than assuming that every query has the average number of matching documents.

Example: Corpus size and matches

At $m = 13$, a 100,000-document corpus has about 12.2 ideal matches on average. Searching only that key usually cannot return the top 100. At one million documents, the average is about 122.1, so one key often contains enough results.

In the independent fixed-coordinate sample, 19 queries had at least 100 ideal matches and one had 99. Equation (70) predicts mean recall $(19 \times 100 + 99)/(20 \times 100) = 99.95\%$. The scan that opened only that cluster measured exactly 99.95%. No fitted embedding-accuracy percentage was needed.

If the strong coordinates move between queries, a fixed first- m key may contain only a few of them. Equation (70) then does not describe that key's recall. The assumption is about the coordinates actually indexed, not simply the number of indexed bits.

9.3 Exact key work

For a key containing exactly the m strong coordinates, equal strong magnitudes make all keys with r mismatches share a prefix penalty. Let Z_r be the number of documents whose strong key has exactly r mismatches. For a fixed query,

$$\Pr(\text{one document has } r \text{ strong mismatches}) = \binom{m}{r} 2^{-m}.$$

Together, these counts follow a multinomial distribution, which describes how the documents are divided among the different mismatch counts. Counts at different radii are not independent. Let r_* be the first radius with at least K documents in total:

$$r_* = \min \left\{ r : \sum_{i=0}^r Z_i \geq K \right\}.$$

The measured exact key search finishes every key with the same penalty at that radius. Under Equation (69), it can then reject all larger radii. Its work is

$$H = \sum_{i=0}^{r_*} \binom{m}{i}, \quad F = \sum_{i=0}^{r_*} Z_i. \quad (72)$$

H includes empty keys. F counts full document scores in the present implementation. For $r_* = 1$ and $m = 13$, that means 14 key attempts, not one attempt and not 2^{13} attempts. A smaller lookup limit invalidates the exact-stop conclusion. A candidate quota can also stop before the layer is complete, so its recall must be measured using the actual stopping rule.

The probability that radius r contains enough documents is another binomial tail:

$$\Pr(r_* \leq r) = \Pr \left(\text{Binomial} \left(N, 2^{-m} \sum_{i=0}^r \binom{m}{i} \right) \geq K \right).$$

This gives probabilities for different amounts of work. The average number of documents in a key does not determine the exact count for every query. No independence assumption between the different radii is needed.

9.4 Split depth

B orders the coordinates using the actual query magnitudes, so it can follow the strong signs even when their positions change. Suppose at least K documents match all m strong signs. After following $j \leq m$ preferred splits, those K correct results are still present. After the remaining set is scored, the full-score bound can reject alternatives that missed a strong bit.

With $W = \lceil N/w \rceil$ physical words per mask, the one-path approximation is

$$T_B(j) \approx T_{\text{prepare}} + jWc_{\text{split}} + Wc_{\text{leaf}} + N2^{-j}c_{\text{score}} + T_{\text{queue}}(j). \quad (73)$$

Both split and leaf loops keep the original width W . For an actual corpus, the observed surviving count replaces $N2^{-j}$. If fewer than K documents match all strong coordinates, more branches may be required and this single-path expression is insufficient; the study retains those queries.

An additional cut saves about $N2^{-(j+1)}$ document scores. It helps only when those saved scores cost more than another full-width split and the added queue work:

$$N2^{-(j+1)}c_{\text{score}} > Wc_{\text{split}} + T_{\text{queue}}(j+1) - T_{\text{queue}}(j). \quad (74)$$

Ignoring the smaller queue term gives a first estimate from the derivative:

$$\frac{dT_B}{dj} = Wc_{\text{split}} - N \ln(2) 2^{-j}c_{\text{score}} = 0, \quad j_c = \log_2 \left(\frac{Nc_{\text{score}} \ln 2}{Wc_{\text{split}}} \right).$$

The positive second derivative shows that the slope of the score term increases with depth. The model requires checking neighboring integer depths and the boundaries 0 and m , then testing actual leaf sizes near $N2^{-j}$. This identifies a choice under the stated model; the derivative alone does not determine the best clustering, query path or hardware behavior.

When N is large, N/W is about the fixed word width w . Thus j_c need not grow with N . A fixed leaf size can be a poor scaling choice: it forces extra cuts as the corpus grows, even when another full-width mask pass costs more than the document scores it removes. In this model, both a global dense-bitplane search and a scan with fixed dimensions still do work that grows at least linearly with N . B can reduce the work per document, but it still requires an initial pass over the full mask.

For $j \geq 1$ with nonempty alternatives, the last split can hold $j+2$ full masks: queued alternatives, the parent and both children. Counting just the mask data, without overhead, the peak is

$$M_{\text{masks,peak}} = (j+2)Wb_w.$$

With no split, only the root mask is needed. Queue metadata, score tables, outputs and the persistent index are additional memory.

9.5 Fixed coordinates

All methods had access to direct-key routing as well as the centroid and sign-prefix scans. There were 704 tuning configurations and 27 independent evaluation workers. The fastest tested key

configuration used one physical cluster and a 13-bit key. The scan baseline chose direct routing into 8,192 clusters, which uses the same 13 strong signs.

At one million documents, C measured 0.0127 ms median latency and 100% binary recall. A measured 0.0145 ms and 99.95% recall. Their paired speedup estimate was 1.145, with a 95% interval of about 1.095 to 1.196. Both met every requested target through 99%. At 100,000 documents and the 99% target, both reached 100% recall: C measured 0.0151 ms and A 0.0186 ms, a paired estimate of 1.227 with interval about 1.144 to 1.364.

The difference is a few microseconds, and the methods largely overlap in candidate selection. C enumerates keys and scores their rows in one C++ call. A ranks cluster keys in one C++ call, then scans their rows in another. The measurements show an advantage for this implementation of combined lookup and scoring. They do not establish an advantage over a scan using the same candidate-selection rule. Larger cluster and key grids may also change the fastest tested setting.

9.6 Changing coordinates

The second setup uses the same document signs, but chooses each query's strong coordinates with every set equally likely and no coordinate repeated. Fixed-prefix keys cannot follow those changing positions. B can, because its coordinate order comes from the query itself.

This run measured 1,001 tuning settings and 60 independent evaluation workers. At one million documents and the 99% requirement, the frozen B configuration used one cluster, unlimited traversal and leaf size 128. It reached 100% binary recall at 0.1496 ms median latency. The scan configuration used 256 centroid clusters and 255 probes, reaching 99.95% at 24.9377 ms. Their paired speedup estimate was about 166.7, with a 95% interval of 147.6 to 183.9. C reached 100% at 26.8429 ms using one cluster and a 12-bit key.

At 100,000 documents, the 99% target comparison reached 100% recall for all three: A measured 2.5438 ms, B 0.1375 ms, and C 2.7170 ms. B's paired speedup over A was about 18.5, with a 95% interval of 18.1 to 19.0. These comparisons use settings selected from the stated grid. Their intervals reflect variation across recorded queries and process blocks. They do not prove that the selected settings are best across every cluster count, leaf size, router or implementation.

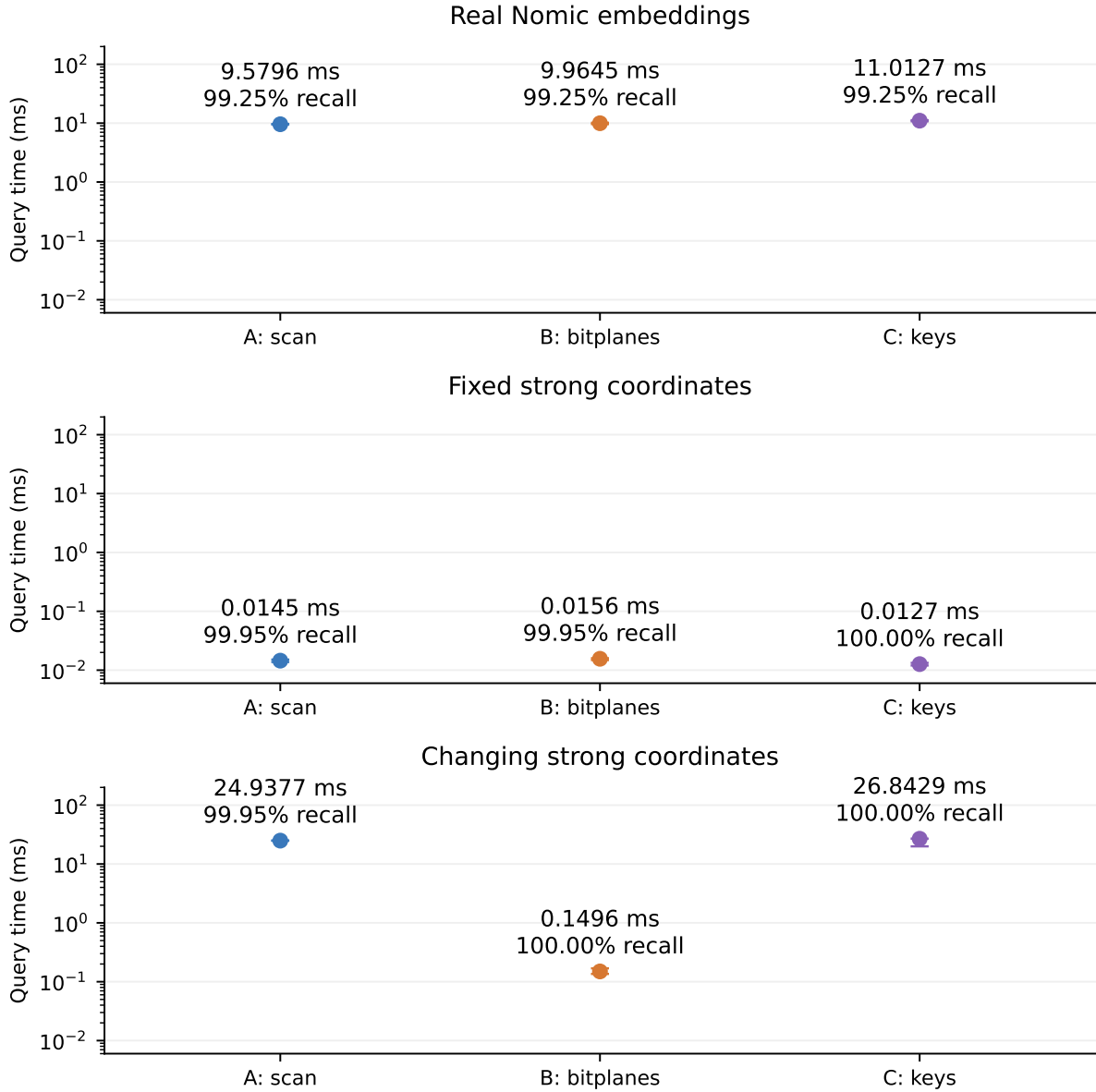


Figure 9: Independent measurements at one million documents, a 99% binary-recall requirement and a 32 GB memory limit. Labels show median time and achieved recall. The vertical axis is logarithmic: adjacent major ticks differ by a factor of ten. The real-data evaluation has 200 queries; each constructed case has 20.

9.7 Predicted work

The count checker uses document signs recorded during preparation, independently of native index traversal. It checks B’s single-path formula only when at least K documents match all strong coordinates and a leaf is reached before splitting weak coordinates. For C, it checks the single-key bound case and Equation (72) when the key contains all strong coordinates. Empty keys are counted, including keys in the final tied layer.

Across the completed tuning and evaluation runs, 447 applicable per-query predictions matched the C++ work counters exactly. This includes split words, leaf words, full scores, key attempts and the tracked B mask peak where applicable. Other queries and clustered configurations remain in the benchmark; the checker reports them as outside these specific prediction conditions.

It does not silently treat them as successful predictions.

The count checks support the formulas under their stated conditions. The timings measure these implementations on the specified data. Neither result alone determines latency for future workloads. In particular, if a new machine changes the relative cost of a bitmap pass and a full score, Equation (74) can favor a different leaf size.

Settings and evidence are in `results/controlled-fixed-2026-09-10/` and `results/controlled-adaptive-2026-09-10/`. The small runnable example is `examples/small_search.py`; it prints one query, its stored document bits, exact results and each method’s work. `experiments/check_controlled.py` connects the conditional formulas to the recorded counters.

10 Complete cost checks

Correct work counts are necessary, but they do not make a latency prediction exact. Two loops can visit the same number of rows while reading very different memory locations. A small candidate set reused repeatedly can stay in cache; different candidates may require fresh memory reads. The following checks distinguish these effects and evaluate complete C++ calls.

10.1 Buffer sizes

The standalone profile uses the shared C++ score table and top- K code. Its path operation allocates a root bitmap, makes both children at each split, keeps the alternatives alive, releases each parent, enumerates the final leaf and releases the remaining masks. The check confirms that the surviving path and alternatives together contain every original document exactly once.

Widths range from 512 bytes to 125 MB. The largest width is one bit over a billion document positions. Thirteen stored planes and the live masks reached about 3.50 GB of process memory. This measurement covers the specified buffers rather than the full 256-plane index. Whole-index memory includes every stored plane, packed document and ID.

The larger buffers showed median split costs of roughly 0.34–0.37 ns per physical word in this standalone loop. At 256 dimensions, contiguous scoring took about 27 ns per document at one and four million rows. Small posting lists cost more per row because fewer rows share the work of preparation and result selection. Compilation and the rest of the query still affect timing; these measured costs do not apply to every CPU instruction or workload.

10.2 Row access cost

A second operation reproduces the leaf-scoring loop: read a bitmap word, find its set bits, fetch those document codes and IDs, score them, and maintain top K . It uses the same scorer as the sequential loop. One run reuses a candidate bitmap after warmup. Another prepares a different bitmap before each measured trial.

16M-document buffer	Reused candidates	Changing candidates
About 1,958 scored rows	0.159 ms	0.455 ms
About 62,500 scored rows	4.400 ms	8.271 ms

Both runs select the same fraction of rows, but select different IDs. Generating candidates happens outside the timer, although it still changes which data the cache already holds. The difference shows why the scan's cost per row should not be used directly for B's scattered leaf scores. It does not directly measure cache-miss counters or establish fixed CPU frequency.

A three-term fit used a fixed cost, physical bitmap words and scored rows. It was fitted on 1M and 4M pools and checked on a 16M pool. The reused-candidate model underestimated one check by 45.2%. With changing candidates, the worst error was still an underestimate of 27.6%. Both fits remain saved. The largest row pool was about 640 MB; it was not a 40 GB row-store test.

The timer for fetching and scoring leaf rows already includes finding their set bits. Adding the separate timer for finding those bits would count that work twice. Also, a fitted fixed cost of zero does not mean score-table preparation is free: the fit may include its cost in other terms.

10.3 Larger C++ runs

The native check uses 1M, 2M and 4M binary documents, then a separate 8M run. It uses the same constructed distribution of strong and weak query coordinates. Packed codes are generated in small chunks and the first million rows are checked against the earlier document pool.

An independent reference avoids expanding these rows into a large float matrix. For each query, it counts strong and weak mismatches directly from the packed bits. With magnitudes a and ϵ , the score is

$$S = a(m - 2e_{\text{strong}}) + \epsilon(d - m - 2e_{\text{weak}}).$$

The reference sorts documents by this score and then ID. Small tests compare that calculation with the earlier direct float sum, including a dimension count that does not fill the final packed word. All complete C++ calls in the scaling check matched the independent reference.

For B, the leaf threshold grows with the corpus: $L = 160N/10^6$. This preserves the intended 13 strong-coordinate cuts. With fixed depth and density, split words, leaf words and scored rows all grow approximately in proportion to N . A simple complete-call model is then

$$\hat{T}(N) = a_0 + b_0N.$$

The fixed term covers work that stays the same as the corpus grows; the slope combines work that grows with it. This model assumes the specified path and growing leaf threshold. It does not justify keeping $L = 128$ while increasing N indefinitely.

10.4 Model corrections

The first fit minimized percentage error over individual queries but was used to predict a median. Fast queries received more weight, which did not match the intended median prediction. For example, workloads with times $[1, 10, 10]$ and $[2, 20, 20]$ have medians 10 and 20. Weighting each query by inverse time favors the fast minority and predicts the wrong median. The corrected fit uses one median per corpus size. A test fails for the mixed workload under the old calculation and stays correct for a uniform workload. The original fit is preserved.

A linear row term was also insufficient for C's small postings. The shared result heap contributes measurable selection cost when only a few hundred documents are scored. Under independently

ordered continuous scores, document i enters the current top K with probability K/i . The expected number of replacements after the first K rows is

$$\sum_{i=K+1}^F \frac{K}{i} \approx K \ln(F/K).$$

For fixed K , a useful approximation is

$$\hat{T}_C(F) = a_C + b_C F + c_C \ln(\max(1, F/K)). \quad (75)$$

Binary ties and key ordering change the exact replacement count, so the logarithm is a model term, not an exact counter. A large contiguous run supplies the per-row cost; the native 1M/2M measurements fit the remaining terms. This corrected the small-posting prediction without changing the reference ranking or search implementation.

The revised 4M check helps diagnose the model, but its results had already been seen. The fitted costs were fixed before the 8M run, which checks predictions on a larger corpus with the same prescribed queries. It does not check performance on new queries.

Strong coordinates	Method	Measured ms	Predicted ms	Error
Fixed	A	197.029	203.016	+3.0%
Fixed	B	0.854	0.884	+3.6%
Fixed	C	0.0481	0.0518	+7.7%
Changing	A	196.664	197.816	+0.6%
Changing	B	0.900	0.859	−4.5%
Changing	C	202.767	207.086	+2.1%

All six checks met the earlier 20% criterion, supporting the stated model over the checked sizes. The largest observed error does not provide a guaranteed error bound at a billion rows. The earlier failed fits remain part of the evidence.

10.5 Leaf-size follow-up

At one million rows, the ideal 13-bit group has mean $\lambda \approx 122$ and standard deviation about 11. A leaf size of 128 can force another cut when the strong-matching group is slightly larger than average. A rough upper margin $\lambda + 3\sqrt{\lambda}$ is about 155; choosing 160 avoids many unnecessary weak-coordinate cuts. The preceding 12-bit group has mean about 244, so it will usually still be split. These are distribution-based settings to check, not universal thresholds or a guarantee from a normal approximation.

The follow-up adds 16,384 clusters, neighboring key widths and leaf sizes, including 160 for the changing-coordinate case. It remeasures prior choices rather than comparing fresh timings against old ones. The first 40 generated queries become the tuning set; query IDs 40–59 are used only after the new choices are frozen. Before a router is reused, the document and reference prefixes are checked against the earlier data.

At 1M rows and the 99% target, the changing-coordinate follow-up selected B with one cluster and $L = 160$. It measured 0.1446 ms at 100% binary recall. A measured 24.9235 ms at 99.95%: a paired speedup estimate of 172.3, with 95% interval 169.1–177.2. C chose a 14-bit key and measured 14.6301 ms at 100%, but its paired speedup interval against A included one. That point estimate alone does not establish a C advantage.

The fixed-coordinate follow-up selected a 16,384-cluster, two-probe scan and a global 13-bit key index. Both reached 100% recall: A measured 0.0140 ms and C 0.0128 ms. Their small timing gap again compares closely related ways to select candidates; it does not prove that one method is always faster.

10.6 Key width and median time

When strong coordinates change with each query, a fixed h -coordinate key contains a random number S_h of the m strong coordinates. The hypergeometric distribution describes this count when coordinates are selected without repetition:

$$\Pr(S_h = s) = \frac{\binom{m}{s} \binom{d-m}{h-s}}{\binom{d}{h}}. \quad (76)$$

For $d = 256$, $m = 13$, the probability of capturing none is 52.75% at $h = 12$, 49.94% at $h = 13$, and 47.27% at $h = 14$. A small width change can move a sample across the halfway point between queries that score nearly all rows and queries that can discard about half.

Assuming the stated gap between strong and weak scores, and also that bounds from weak key coordinates cannot rule out individual weak patterns, the work for a query capturing s strong coordinates is approximately

$$H = 2^{h-s}, \quad F = N2^{-s}.$$

The expected scored fraction is $\sum_s \Pr(S_h = s)2^{-s}$: about 71.30% at $h = 13$ and 69.44% at $h = 14$. The mean work changes smoothly even when a small sample's median changes sharply. The analysis therefore requires retaining uncertainty across queries and the full range of query times. The condition is about the stated constructed distribution, not arbitrary embeddings or IVF.

10.7 One-bit control

The original real-data parameter grid started with four-bit keys. A final control gives C a one-bit key and A's routing layout, with no candidate or lookup limit. It keeps A and B's frozen settings and uses 100 further query IDs, excluding all 500 previously used IDs. No parameter is changed based on this control's outcomes.

A measured 9.541 ms, B 9.855 ms and C 10.545 ms. All three reached 98.95% binary recall, narrowly below the requested 99%. The report retains this miss without adding a tolerance to classify it as a pass. There is no supported alternative speedup in this check. The original 200-query evaluation remains a separate result, where its conservative choices met their sample targets. A finite-sample selection rule does not guarantee the same mean recall on every later query sample.

10.8 Text-query timing

The final check runs actual text through the cached local Nomic model using Apple's MPS backend, then applies the same normalization, routes the resulting vector and calls the C++ search. Batch size is one. Each sequence shape is warmed first; model loading, index building and reference creation are outside the request timer. References are computed afterward from the actual encoded vectors.

Method	Encode median	Retrieve median	Total median	Total p95
A	6.488 ms	9.406 ms	16.313 ms	19.201 ms
B	6.500 ms	9.651 ms	16.303 ms	19.217 ms
C	7.543 ms	11.201 ms	18.912 ms	21.898 ms

These are 20 existing independent-query IDs, three blocks and two repeats, with the earlier frozen settings. All reached 99.45% binary recall on this sample. The small A/B difference does not establish a speedup. Each total is measured per request: its median is not the sum of the two stage medians. The process lifetime peak was about 3.67 GB including reference construction. The MPS driver reported about 1.08 GB allocated. Some of that may already be included in the process memory figure (RSS), so adding the two values may double-count memory.

This text check applies to the real Nomic queries. The constructed strong-coordinate vectors do not come from this encoder, so adding its time to those examples would not constitute a measured end-to-end sparse-embedding pipeline.

11 Method comparison

The three methods share query embedding, preparation, routing, filtering, and the optional final reranker. Their difference is the work used to obtain candidates from the selected pool. Explicit shared costs permit comparisons of both index work and complete query time.

11.1 Query time

Using the earlier definitions,

$$T_A = T_{\text{common}} + P c_{\text{range}} + N_{\text{elig}} G c_{\text{lut}} + H_s(N_{\text{elig}}, R) c_{\text{heap}} + T_{\text{rank}}(R) + T_{\text{finish}}(R'), \quad (77)$$

$$T_B = T_{\text{common}} + T_{\text{order}} + T_{\text{roots}} + \sum_j s_j W_j c_{\text{split}} + \sum_j \ell_j W_j c_{\text{leafword}} \\ + V c_{\text{node}} + F G c_{\text{lut,leaf}} + H_s(F, R) c_{\text{heap}} + T_{\text{rank}}(R) + T_{\text{finish}}(R'), \quad (78)$$

$$T_C = T_{\text{common}} + T_{\text{keyprep}} + E c_{\text{enum}} + C^{\text{copy}} c_{\text{copy}} + \sum_j (H_j c_{\text{lookup}} + F_j c_{\text{post}}) \\ + F G c_{\text{lut,posting}} + H_s(F, R) c_{\text{heap}} + T_{\text{rank}}(R) + T_{\text{finish}}(R'). \quad (79)$$

The expression for C follows the implemented shared stream: E counts its enumeration work once, and C^{copy} counts its mask-copy work once. Cluster sums cover actual directory lookups and postings. $F = \sum_j F_j$, and the current scorer evaluates all G groups per candidate. A tail-only or full-key score-reuse implementation would substitute its actual group count; those savings are not credited to the measurements here.

11.2 End-to-end speedup

If two configurations have identical embedding and final-stage costs, write their shared time as T_0 . Replacing index time t_A by t_B gives speedup

$$\frac{T_0 + t_A}{T_0 + t_B},$$

not t_A/t_B . Even removing index time entirely cannot exceed $1 + t_A/T_0$ when $T_0 > 0$. If candidate counts or final-vector fetches differ, those costs are no longer shared and must be evaluated separately. Cached-query index timings and complete text-query timings therefore require separate reporting.

11.3 Stored index memory

The earlier layouts count the bytes required for the data itself. The implemented search cores also reserve space; their allocated memory is therefore counted explicitly:

$$M_{A,\text{core}} = M_{\text{codes,cap}} + M_{\text{IDs,cap}} + M_{\text{other,A}}, \quad (80)$$

$$M_{B,\text{core}} = M_{\text{codes,cap}} + M_{\text{IDs,cap}} + M_{\text{planes,cap}} + M_{\text{other,B}}, \quad (81)$$

$$M_{C,\text{core}} = M_{\text{codes,cap}} + M_{\text{IDs,cap}} + \sum_j (U_j b_{\text{node}} + L_j b_{\text{ptr}}) + M_{\text{other,C}}. \quad (82)$$

“cap” means allocated array capacity. The remaining terms cover bucket objects, the outer label map and allocator bookkeeping not already counted. C’s IDs are its postings; there is no second N -row posting permutation. Its hash-node and bucket-pointer terms use the separate chaining layout in Equation (67).

The data stored in the planes takes $db_w \sum_j \lceil n_j/w \rceil$; allocated capacity is at least that amount. Storage sums run over all clusters, not just the probed ones. Add any retained routing index, filters and fine-ranking vectors to the C++ index to obtain the index memory kept loaded during search. Input arrays kept only during construction belong to the peak while building, not the memory kept during search.

11.4 Peak query memory

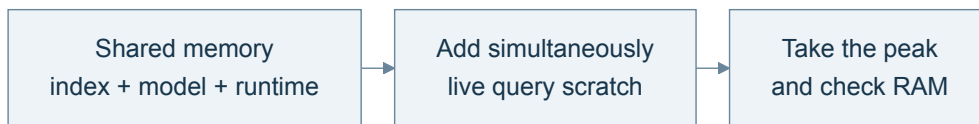


Figure 10: Count stored index data, runtime reserve, and temporary memory used at the same time separately. Check when each stage allocates and releases its buffers.

The temporary memory used during search is

$$Q_{A,\text{search}} = Q_{\text{base}} + Q_{\text{scan workspace}}, \quad (83)$$

$$Q_{B,\text{search}} = Q_{\text{base}} + Q_{\text{order}} + \max_t [\gamma_B Q_{B,\text{frontier}}(t) + Q_{\text{current}}(t) + Q_{\text{children}}(t)], \quad (84)$$

$$Q_{C,\text{search}} = Q_{\text{base}} + Q_{\text{keyorder}} + Q_{\text{selected}} + \max_t [\gamma_C Q_{\text{enum}}(t) b_{\text{state}}]. \quad (85)$$

C has one queue for the common key coordinates, not one queue per probed cluster. Q_{keyorder} stores the sorted coordinate indices; Q_{selected} stores the selected bucket pointers. $\gamma_C Q_{\text{enum}}$ accounts for queue capacity rather than only live entries. The candidate heap is already in Q_{base} , so it is not added again. The current C implementation reads posting ranges directly and allocates no separate gathered-document buffer.

At the Python boundary, native result vectors and output arrays may coexist. Include that interface workspace when evaluating the full request peak. Whole-process observations include it even when individual interface allocations are not separately measured.

Small iterator state is included in workspace terms. Embedding and final reranking can have more temporary memory than search, so use Equation (12) to obtain the full query peak. With loaded index size $M_{m,\text{resident}}$ for method m , the RAM constraint is

$$M_{m,\text{resident}} + M_{\text{model}} + M_{\text{runtime}} + \max_t \sum_{u \in \mathcal{A}(t)} Q_{m,u}(t) \leq M_{\text{RAM}}. \quad (86)$$

M_{model} is the shared memory for the loaded embedding model if encoding runs locally; it is zero in the memory being counted here when an external service performs encoding. This shared term is not multiplied by the number of queries.

For a conservative bound, replace the sum by $JQ_{m,\text{peak},\text{max}}$. A loose upper bound that exceeds RAM does not prove that a tighter implementation cannot fit. An observed peak below the limit does not prove that every possible query fits. Each observation supports a different claim.

The native measurements use the allocation choices in Equation (67): C reorders codes and IDs together, so its extra posting permutation is absent. Its directory uses allocated nodes and bucket pointers. Use that layout when connecting the general formulas to the measured results.

11.5 Method characteristics

	A: full scan	B: bitplanes	C: key enumeration
Additional index	none beyond common layout	$db_w \sum_j W_j$	key directory; IDs already counted
Main work	every eligible code	split masks, leaf masks, full scores, pending nodes	empty and occupied keys, enumeration, postings, full scores
Variable counts	N_{elig}	s_j, ℓ_j, f_j, V	$H_j, F_j, E, Q_{\text{enum}}^{\text{max}}$
Main memory concern	selection or batch score buffers	many full-width pending masks	pending flip sets and interface outputs
When local results are exact	scan all eligible rows	valid bounds, sufficient traversal, correct ties	full-key weighted order or a valid full-score bound
Settings that affect recall	routing probes; candidate/rerank boundary	node budget, visit policy, earlier routing	prefix width, candidate or key budget, earlier routing
What must be measured	scoring cost, routing quality	skipped scores versus bitmap/queue cost and recall	number of empty keys versus useful candidates, timing and recall

11.6 Cost sensitivity

Documents per physical cluster, n_j . A grows roughly linearly with eligible row count. In B, a larger cluster also widens every active bitmap. In C, more documents per key can reduce the number of empty probes needed, while increasing posting and scoring work.

Binary dimension, d . Longer codes increase score terms and stored bits. B may gain more from avoiding full scores, but also stores more planes and can traverse more coordinates. For full-key C, the number of possible keys grows exponentially. For short-key C, h and d are separate parameters.

Probes, P . More clusters can recover missed neighbors. They also increase eligible rows, root masks, active key streams and candidate competition. Cluster sizes and the query’s routing behavior matter, not just the number P .

Leaf size and node budget. A small leaf threshold can trade scoring for more mask work. A finite node budget caps visits but does not specify how many useful leaves are reached. The best choice depends on the query and data; recall need not change in one direction as these settings vary.

Lookup width and candidate target. A short key has denser postings and a smaller key space, but leaves more score information out. A longer key can improve selection of useful candidates and simultaneously make empty-key enumeration costly. A strong prefix can improve recall; that does not remove the enumeration cost.

Scoring code and memory system. SIMD, whether data stays in the CPU cache, batch size, alignment, layout and thread count change the effective coefficients. Keep them fixed for an algorithm comparison, or report their changes as a separate experiment.

11.7 Constrained configuration selection

Let θ_m contain the chosen method’s tunable parameters and let r_0 be a recall target. The objective uses a specified latency measure, such as the median or p95 across queries:

$$\theta_m^* \in \arg \min_{\theta_m} \text{Quantile}_p[T_m(q; \theta_m)] \quad (87)$$

subject to the RAM condition in Equation (86) and

$$\mathbb{E}_q[\text{Recall}@K(q; \theta_m)] \geq r_0. \quad (88)$$

The expectation specifies an average-recall target. A requirement that almost every query reach r_0 is stronger and needs a different probability constraint.

Listing 14: A defensible finite-grid experiment.

```

ChooseConfiguration(corpus, queries, fixed_reference, RAM_limit, targets):
    build or cache every configuration's required index
    compute exact reference results once for each query
    on tuning queries:
        for every configuration in the declared grid:
            measure latency, quality, work counts and peak RAM
            keep failures and empty returns in the record
        choose qualifying configurations for each target
    on separate test queries:
        evaluate those choices without retuning
    report lowest tested latency only where the target and RAM checks pass
    distinguish measured results from untested model predictions

```

The parameters are often integers: cluster count, probes, lookup bits, and visited nodes. A small declared set of combinations handles these choices directly. Differentiation can simplify a smooth part of the cost model, but it does not determine whether the recall constraint is satisfied. For example, a model $T(C) = aC + bNP/C$ has a stationary point, where its derivative is zero, at $C = \sqrt{bNP/a}$ when $a, b > 0$ and P is held fixed. Check nearby integer values that meet the constraints, including their memory use. If the probes required for the recall target change with C , that derivative no longer solves the complete problem.

Configuration selection uses fitting and tuning queries, followed by evaluation of the fixed choice on separate queries. Optimizing an inaccurate recall model can select the configuration where that model makes its most optimistic error. At 99% recall, a predicted 99% is insufficient; report actual query-level quality and its uncertainty. The existing normal prediction's real-data errors are too large to establish the fastest qualifying configuration.

If d is also varied, keep the reference used to judge quality fixed across choices, for example the same long-vector top K or the same relevance labels. Recomputing a different binary ground truth for every d can make a small representation appear perfectly accurate against a changed reference task.

11.8 Billion-document estimates

The C++ cost models were checked through 8M rows. The bitmap test reaches the full width of a 1B-row mask, but no complete 1B index was built or searched. The numbers below are estimates under the same 256-bit score, top 100, one CPU thread and stated distribution of strong coordinates. A different machine or pattern of memory reads can change the costs.

Memory requirements can rule out a configuration before latency is estimated. For the current layout at $N = 10^9$ and $d = 256$,

$$M_{\text{codes}} = 32 \text{ GB}, \quad M_{\text{IDs}} = 8 \text{ GB}.$$

A and C need at least 40 GB before directories, containers and query buffers. B retains a second transposed copy, adding at least 32 GB. Therefore all three fail a 32 GB limit, and B also fails a 64 GB limit. Changing probes does not remove stored documents. These rejections apply to the current in-memory layouts; they are not claims about every compressed or disk-based design.

The planning figures below allow room for up to twice the stored row-ID count, a small allowance

for directories and containers, and a configurable 1 GB reserve for running queries. That reserve is assumed, not measured. Payload counts exclude overhead: if even those bytes exceed the budget, rejection is certain for this layout. A configuration that fits the planned budget still requires memory validation. Building an index from full input copies can require more RAM than serving it.

1B-row path	Payload lower bound	Planned query RAM	64 GB budget
A, known strong-prefix routing	40 GB	about 49 GB	Modeled fit
C, global 13-bit key	40 GB	about 49 GB	Modeled fit
B, one global cluster	72 GB	about 83 GB	Rejected
B, same prefix route, scan-like leaf	72 GB	about 81 GB	Rejected

All these planning figures fit a 1 TB budget. The model does not claim that a 1 TB machine has the M5 Max’s exact memory behavior. The measured row buffers were much smaller than a 40 GB store.

11.9 Fixed strong prefix

With thirteen fixed strong coordinates, the ideal group contains an expected $10^9/2^{13} \approx 122,070$ rows. Under the independent-bit assumption, a binomial calculation describes the chance of having fewer than 100. A Chernoff bound puts that probability below 10^{-26464} in this example. This extremely small bound belongs to the stated distribution; it is not an embedding-accuracy guarantee.

A can route directly to that group and scan it. C can look up the same key and score its posting range. Their modeled costs are both roughly 3 ms. The data being read after routing is only a small part of the corpus, although all codes and IDs are still stored.

The comparison gives B access to the same prefix routing and a leaf large enough to scan the selected group. An all-ones leaf-mask profile shows a similar scoring cost. Thus B’s scan fallback is also estimated at roughly 3 ms, while storing a much larger index. Requiring B to traverse the entire billion-row bitmap would deny it this candidate-selection option. The model is not precise enough to distinguish a timing gap of a few percent here. A or C needs less memory, but the estimate does not prove one method is always faster.

11.10 Changing strong coordinates

A fixed prefix no longer supplies the same filter. The measured B path can select its columns from the actual query. Keeping thirteen cuts and scaling the leaf threshold to 160,000 gives an expected final set of about 122,070 rows. Its mask payload at the last split is

$$(13 + 2) \left\lceil 10^9/64 \right\rceil \times 8 = 1.875 \text{ GB.}$$

The model fitted to C++ runs estimates about 0.10 s for this global B path. Doubling the part of its cost that grows with corpus size raises the estimate to about 0.20 s. This checks the effect of a higher assumed cost; it is not a confidence interval or promised bound. The model estimates about 25 s for a global scan.

That comparison does not establish that B beats every possible IVF configuration. Let f be the fraction of the corpus A actually scores, and let T_{route} be the time needed to obtain those

candidates. A’s approximate cost is

$$T_A \approx T_{\text{route}} + fNc_{\text{scan}}.$$

Ignoring routing time gives A the most favorable comparison. At the current cost estimates, its scanned fraction must be below about 0.41% to undercut the 0.10 s B point, or below about 0.82% if B’s variable cost doubles. It must still meet the recall target at that fraction. Scanning can be faster if routing produces a sufficiently small, accurate candidate pool at low cost. Otherwise, a global or less heavily clustered B path warrants evaluation.

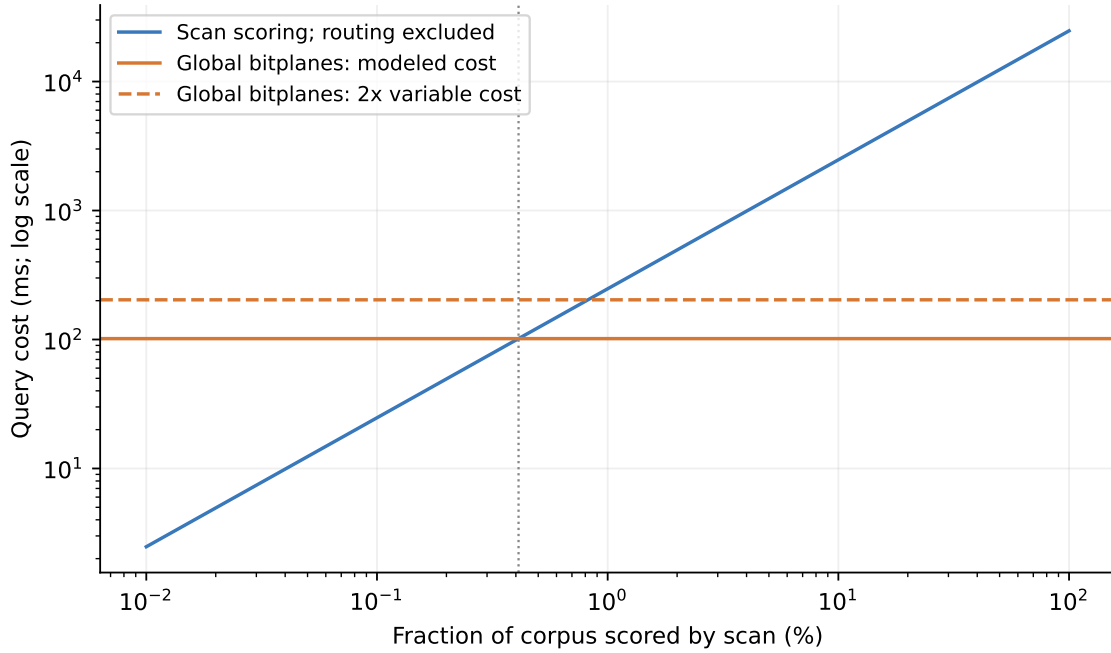


Figure 11: A conditional break-even calculation at 1B rows with sufficient RAM. The scan line excludes routing time, favoring A. B is one fixed global operating point, not a retuned curve. Routing recall and the concentrated-query assumption must be checked separately.

B can also use a router. Its cost then uses the physical cluster widths and visited work in the earlier sum over clusters. The graph does not solve that larger optimization problem or assume a relevant document is always in the probed clusters.

For C, the number of strong coordinates actually captured by its key matters. Under Equation (76) and the stated bound conditions, a query capturing none can still require scoring nearly all rows; capturing one leaves roughly half. A fixed short key can therefore remain expensive even when it has many occupied entries and few empty lookups. A low cost for finding a nonempty key does not establish low total search cost.

11.11 Scope of the findings

The completed tests establish three outcomes within their stated experimental scope:

- On the real Nomic sample, no alternative speedup was established. Even a one-bit key control did not improve latency. Recall targets must retain their sampling uncertainty.
- With a fixed strong prefix, direct routing and integrated key lookup are closely related and

both avoid most scoring. C had a small measured implementation advantage in the controlled runs.

- With changing strong coordinates, B's query-dependent column order produced a large, independently checked advantage in the constructed data. Increasing the leaf size with corpus size preserved a useful path; keeping a tiny fixed leaf would not.

At a billion rows, the minimum memory counts are exact for this layout. Query times and the best routing settings remain estimates under stated assumptions. The formulas give settings to test and the conditions under which costs would be equal. A finite parameter grid, or an embedding model name, cannot identify the best setting for every workload.

The current calculation is implemented in `experiments/project_costs.py`; its inputs and all scenarios are saved in `results/projected-cases-2026-09-11/`. The earlier `reports/search-math/numerical-cases/calculate.py` is preserved as a historical assumed-cost model. Its extra-posting and hash-slot layout must not be substituted for the measured native C layout.

Comparison findings

A scores the selected pool. B identifies scores that can be avoided, with dense bitmap operations potentially accounting for substantial cost. C identifies keys containing useful candidates, incurring costs for empty keys and queue operations as well. Each can be exact under a suitable stopping rule. Which is fastest at a recall target is determined by the data, the representation, the visited work, the memory layout and the hardware. The exact small examples explain how recall can be calculated. The measured prediction errors show where the assumptions need more evidence before they can guide a real configuration choice.

A Expected counts by distance

The count $G_K(p)$ in Equation (46) can be evaluated directly as

$$G_K(p) = \sum_{x=0}^N \min(K, x) \binom{N}{x} p^x (1-p)^{N-x}.$$

For large N , computing the binomial coefficients and powers separately can overflow or lose small probabilities. A shorter expression follows from a counting identity. For an integer $x \geq 0$, $\min(K, x)$ counts how many thresholds $1, 2, \dots, K$ it reaches:

$$\min(K, x) = \sum_{j=1}^K \mathbf{1}[x \geq j].$$

Taking expectations on both sides gives

$$G_K(p) = \sum_{j=1}^K \Pr(X \geq j), \quad X \sim \text{Binomial}(N, p). \quad (89)$$

A binomial survival function computes the probability of reaching a threshold; its implementation must preserve small probabilities. The small-example checker uses the direct sum, while the simulation checker uses this shorter expression. Comparing them checks the result with two different calculations.

For the distance model, $\sum_m w_m = G_K(1) - G_K(0) = K$. Thus the weights in Equation (47), divided by K , form a probability distribution over the distance of a uniformly chosen true top- K row. Given that row's distance, symmetry makes its error positions uniform. The ID rule chooses among tied rows without favoring one set of error positions. This justifies multiplying w_m by the hypergeometric survival probability.

The argument would change if ties deliberately favored matching prefixes. It would also change if candidate collection stopped after a corpus-dependent quota. These are changes to the selection policy, not numerical errors in the fixed-radius formula.

Listing 15: Expected top-K recall for a fixed prefix radius.

```
ExpectedTopKRecall(N, d, h, radius, K):
    previous_slots = 0
    expected_found = 0
    for distance in 0 .. d:
        p = F_d(distance)
        slots = sum(Pr(Binomial(N, p) >= j) for j in 1 .. K)
        rows_at_distance = slots - previous_slots
        expected_found += rows_at_distance * g_h_radius(distance)
        previous_slots = slots
    return expected_found / K
```

B Finite-sample prefix recall

Prefix recall can also be calculated exactly for a finite collection, rather than approximated from the score tails. The finite- N_* calculation below assumes document score pairs are independent, follow the same distribution, and have no ties. It shows why correlation alone is not enough.

For a particular document with full score s and prefix score t , classify another document into four events:

$$\begin{aligned} p_{11}(s, t) &= \Pr(S > s, T > t), & p_{10}(s, t) &= \Pr(S > s, T \leq t), \\ p_{01}(s, t) &= \Pr(S \leq s, T > t), & p_{00}(s, t) &= \Pr(S \leq s, T \leq t). \end{aligned}$$

Among the other $N_* - 1$ documents, let (a, b, c, e) count these categories. A multinomial distribution counts how many independent documents fall in each of the four categories, using those probabilities. The chosen document belongs to the full top K when $a + b \leq K - 1$, and belongs to the prefix top R when $a + c \leq R - 1$. Therefore, with $e = N_* - 1 - a - b - c$,

$$P_{K,R}(s, t) = \sum_{\substack{a, b, c \geq 0; \\ a+b \leq K-1; \\ a+c \leq R-1 \\ a+b+c \leq N_*-1}} \frac{(N_* - 1)!}{a!b!c!e!} p_{11}^a p_{10}^b p_{01}^c p_{00}^e. \quad (90)$$

If $F_{S,T}$ is the joint distribution of one document's scores, every document has the same event probability under the identical-sampling assumption. Summing over all N_* documents and dividing by K gives

$$\mathbb{E}[\text{Recall@ } K] = \frac{N_*}{K} \int P_{K,R}(s, t) dF_{S,T}(s, t). \quad (91)$$

This assumes the candidate stage returns exactly the prefix top R and the final stage uses the same full score. Retrieving whole tied prefix keys changes that candidate rule; ties can be handled by also ordering on document IDs, or by including the probabilities of selecting tied documents in the calculation.

Equation (91) is much harder to evaluate than the conditional-normal approximation. It also requires $F_{S,T}$. Matryoshka training, a code length, a cluster count and a RAM budget do not uniquely specify that distribution. An exact formula identifies the required inputs, but it cannot determine them from an algorithm name.

C Implementation checks

1. Fix the score and tie rule. Keep original-float, truncated-float, binary-score and human relevance references distinct.
2. Check the score identity on small vectors and enumerate the actual top K .
3. Check that the weighted-subset stream emits every subset once, from lowest to highest penalty, and processes the final boundary tie before claiming exactness.
4. Count split words and leaf words separately. Use each cluster's physical width, including padding; do not substitute the number of active documents for the bitmap length.
5. Record the ownership of codes, external IDs, posting references and key storage. Any shared or reordered array must be stated before removing its bytes from a formula.
6. Count temporary query memory at the point when the most is held at once. A bound on queue capacity is different from a measured peak allocation.
7. Keep key attempts separate from occupied keys and returned postings. Expected first hit is not the same threshold as finding R candidates.

8. Count the cost of centroid float calculations separately from binary score lookups. Include result selection, memory allocation, copying and fetching scattered rows where they occur.
9. For each block, use either one measured combined cost or separate costs for its operations. Do not count the same memory and selection work in both.
10. Test statistical predictions on queries not used to estimate their parameters. Evaluate recall after the actual candidate and tie rules, not merely a probability of finding some key.

D Rounding and exactness

The score and bound proofs assume arithmetic without rounding. In code, rounding must not lower an upper bound enough to discard a possible result. The local C++ reference includes a rounding allowance for bound pruning and a deterministic row-ID tie order. A floating-point comparison that looks like the formula is not by itself evidence of exactness.

Validation requires checking the implemented arithmetic and comparing its output with a full reference scan on the same inputs. Additional corrections require evidence of a failure. The small examples and archived candidate checks provide validation evidence for their stated scopes; they do not prove every future implementation correct.

E Encoder cost details

The main formulas keep T_{emb} and Q_{emb} as measured model-specific functions. If a more detailed encoder estimate is needed, its architecture must be another set of inputs, not an unexplained constant.

For an illustrative standard dense Transformer encoder, let n_t be query tokens, L_e the number of layers, m hidden width, f feed-forward width, and H_a attention heads. For one layer with ordinary two-matrix feed-forward computation, multiply-add contributions are

$$U_{\text{layer}} = 4n_t m^2 + 2n_t^2 m + 2n_t m f. \quad (92)$$

The first term comes from the query, key, value and attention-output projections. The second comes from attention-score and attention-value matrix products. The third comes from the feed-forward expansion and contraction. Activation, normalization and softmax operations add separate elementwise terms. A different block, gated feed-forward network or sparse attention requires different counts.

The resulting time model is

$$T_{\text{emb}} = T_{\text{tokenize}} + L_e (U_{\text{layer}} c_{\text{enc,mac}} + U_{\text{elementwise}} c_{\text{enc,elt}}) + T_{\text{pool/project}}.$$

These effective coefficients depend on batching, device kernels, precision and memory behavior. The resource bound in Equation (19) provides an alternative; adding it to coefficients that already include the same work would count that work twice.

When attention operations run separately, they may store $H_a n_t^2$ attention values. Scratch means temporary memory used while processing a query. With bytes per activation b_a , the temporary-memory model is

$$Q_{\text{emb}} = b_a (\eta_m n_t m + \eta_f n_t f + \eta_a H_a n_t^2) + Q_{\text{kernel}},$$

where the η coefficients count buffers held at the same time. Fused attention combines operations and can avoid storing the square attention matrix, so it uses different buffer counts. Layer workspaces can be reused during inference; multiplying a per-layer peak by L_e is not a general inference-RAM formula. Resident model weights are shared and counted separately.

A Matryoshka prefix normally reuses an already computed embedding. Halving the search prefix d does not automatically halve the encoder’s layer count or hidden width. It reduces the subsequent representation and search work unless the actual encoder implementation also changes.

F Reproduction and inputs

The editable source is `report.tex`, with chapter files in `sections/`. Mermaid sources and rendered diagrams are in `figures/`. The build instructions are in the adjacent `README.md`. The arithmetic checks are implemented in `verify_math.py`. They validate the worked examples, exact subset ordering, binomial conditional-recall example, word widths, selected storage identities, and quoted measured rows. The statistical evidence is preserved with its source hash.

The table lists inputs a future calculator must obtain. Document count alone cannot supply them.

Input family	Parameters or counts	How to obtain them
Representation	D, d, d_r, d_f , preparation and score rule	fixed experimental definition; embedding/model metadata
Retrieval target	K, R , filter, reference score, tie rule	fixed task definition
Cluster layout	$C, n_j, \mathcal{P}(q), e_j$	built index and per-query routing/filter traces
Storage widths	$w, b_w, b_{id}, b_o, b_c, b_f, b_s$	actual types, packing and alignment
Baseline scoring	g, G, H_s, c_{lut}	source code, work counters and kernel measurements
Bitplane search	$L, B_{node}, s_j, \ell_j, f_j, V$	configuration plus query traces; not budget alone
Bitplane RAM	frontier masks, current/child buffers, capacities	allocation/capacity measurements or an explicit bound
Key index	$I_h, h_j, U_j, \alpha, b_k, b_{ref}$	chosen lookup coordinates, index build and slot layout
Key search	$H_j, F_j, E_j, Q_j^{\max}$, stopping rule	actual key enumeration, posting and frontier traces
Prefix-trie variant	Z_j, b_{trie}, u_j , posting ranges	actual stored nodes, path and range visits
Hardware	compute rates, bandwidth, cache regime, SIMD, threads	calibrated for the actual workload and machine
Optional final stage	fetch policy, R', u_f, d_f , rerank score	execution pipeline and storage-residency decision
Statistical model	query/data distribution, covariance, score tails	declared synthetic assumptions or measured and checked estimates
Serving RAM	$M_{resident}, M_{model}, M_{runtime}, J, Q_{peak}$	ownership/lifetime analysis and process measurements

A numerical example can assign values to these parameters. The equations themselves remain

parameterized, so changing the workload changes the inputs instead of requiring a different explanation of the algorithm.

References

- [1] Exa Team. *How we built a web-scale vector database*. 17 December 2024.
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<https://arxiv.org/abs/2205.13147>.
- [3] M. Norouzi, A. Punjani, D. J. Fleet. *Fast Exact Search in Hamming Space with Multi-Index Hashing*. 2013, revised 2014. <https://arxiv.org/abs/1307.2982>. This concerns Hamming search; it is not a weighted-search guarantee for the methods here.
- [4] Local project evidence. `cpp/index.cpp`, `cpp/index.hpp`, `cpp/key_index.cpp`, `experiments/`, `results/scaling-msmarco-1m/`, `results/fiqa-2026-09-09-dense/`, and `calculator/recall-validation.json`. Reproduction and source hashes are described in Section 7.